

## MambaIRv2: Attentive State Space Restoration

Hang Guo<sup>1,\*</sup> Yong Guo<sup>2,\*</sup> Yaohua Zha<sup>1,6</sup> Yulun Zhang<sup>3</sup> Wenbo Li<sup>4</sup>  
 Tao Dai<sup>5,†</sup> Shu-Tao Xia<sup>1,6</sup> Yawei Li<sup>7</sup>

<sup>1</sup>Tsinghua University <sup>2</sup>Max Planck Institute for Informatics

<sup>3</sup>Shanghai Jiao Tong University <sup>4</sup>The Chinese University of Hong Kong

<sup>5</sup>Shenzhen University <sup>6</sup>Peng Cheng Laboratory <sup>7</sup>ETH Zürich

### Abstract

The Mamba-based image restoration backbones have recently demonstrated significant potential in balancing global reception and computational efficiency. However, the inherent causal modeling limitation of Mamba, where each token depends solely on its predecessors in the scanned sequence, restricts the full utilization of pixels across the image and thus presents new challenges in image restoration. In this work, we propose MambaIRv2, which equips Mamba with the non-causal modeling ability similar to ViTs to reach the attentive state space restoration model. Specifically, the proposed attentive state-space equation allows to attend beyond the scanned sequence and facilitate image unfolding with just one single scan. Moreover, we further introduce a semantic-guided neighboring mechanism to encourage interaction between distant but similar pixels. Extensive experiments show our MambaIRv2 outperforms SRFormer by **even 0.35dB PSNR** for lightweight SR even with **9.3% less parameters** and suppresses HAT on classic SR by **up to 0.29dB**. Code is available at <https://github.com/csguoh/MambaIR>.

### 1. Introduction

Image restoration aims to recover high-quality images from low-quality observations, tackling various sub-problems such as image super-resolution, image denoising, and JPEG compression reduction, and others. With the advent of deep learning, state-of-the-art performance has been consistently achieved. Early works primarily utilized convolutional neural networks (CNNs) as the backbone [12, 13, 30, 45, 53]. Later, vision transformers (ViTs) [15] gained popularity for their superior performance [5, 7, 10, 28, 29]. More recently, the selective state-space model (Mamba) [17] has been explored, showing considerable potential as an alternative backbone for image restoration tasks [18, 36].

\*Equal contribution

†Corresponding author

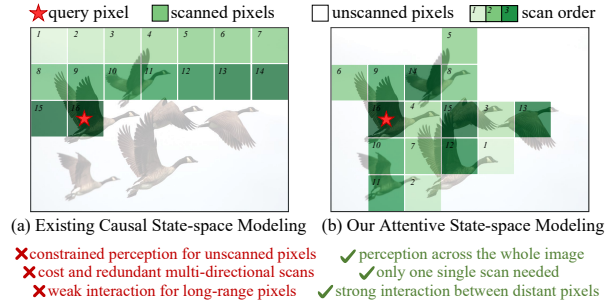


Figure 1. (a) The existing method [18] suffers from the adverse effects of the causal nature of Mamba (the multi-directional scans are not shown for presentation clarity). (b) The proposed MambaIRv2 can achieve attentive state-space modeling that embeds ViT-like non-causal properties into Mamba.

Despite its potential, existing Mamba-based methods face significant challenges, particularly due to their reliance on causal state-space modeling. Specifically, existing methods [18] unfold the 2D image with a predefined scanning rule to generate the 1D token sequence. However, in Mamba, each pixel is modeled based solely on its preceding pixels in the scanned sequence, *i.e.*, the causal property, which results in several detrimental effects for non-causal image restoration tasks. **First**, as shown in Fig. 1(a), the query pixel can only capture information from its preceding ones and cannot perceive subsequent pixels, which results in under-utilization of helpful pixels across the image. **Second**, the inherent causal property leads to the necessity of multi-directional scans, which is widely adopted by existing approaches [18, 36, 39] for mitigating information loss. Yet, this multi-scanning inevitably increases the computational complexity, particularly for high-resolution inputs. Furthermore, an empirical investigation in Sec. 3 reveals that there is also substantial information redundancy among these multi-directional scans. **Third**, our findings in Sec. 3 demonstrate that Mamba [17] is prone to long-range decay in token interaction, meaning distant tokens in the sequence have diminished interactions. Consequently, even previously scanned pixels that are distant yet relevant cannot be effectively utilized by the query pixel.

## MambaIRv2: 注意力状态空间恢复

韩国<sup>1,\*</sup>郭勇<sup>2,\*</sup>查耀华<sup>1,6</sup>张玉伦<sup>3</sup>李文波<sup>4</sup>

陶岱<sup>5</sup>, †夏树涛<sup>1,6</sup>李亚伟<sup>7</sup>

<sup>1</sup>清华大学<sup>2</sup>马克斯·普朗克信息学研究所

<sup>3</sup>上海交通大学<sup>4</sup>香港中文大学

<sup>5</sup>深圳大学<sup>6</sup>彭程实验室<sup>7</sup>ETH Zurich

### 摘要

基于Mamba的图像修复骨干网络近期在平衡全局接收与计算效率方面展现出显著潜力。然而，Mamba固有的因果建模局限性——每个标记仅依赖于扫描序列中的前序标记——限制了对图像中像素的充分利用，从而给图像修复带来新挑战。本研究提出MambaIRv2框架，通过赋予Mamba类似ViTs的非因果建模能力，使其达到注意力状态空间修复模型的水平。具体而言，所提出的注意力状态空间方程可实现超越扫描序列的注意力机制，仅需单次扫描即可促进图像解卷积。此外，我们进一步引入语义引导邻域机制，以促进远距离相似像素间的交互作用。大量实验表明，MambaIRv2在轻量级超分辨率任务中PSNR 优于SRFormer高达0.35dB，且参数量减少9.3%，同时在经典超分辨率任务中将HAT抑制幅度最高达0.29dB。代码可在<https://github.com/csguoh/MambaIR>获取。

### 1. 引言

图像修复旨在从低质量观测中恢复高质量图像，涉及图像超分辨率、图像去噪、JPEG压缩率降低等多种子问题。随着深度学习的兴起，持续取得前沿性能表现。早期研究主要采用卷积神经网络（CNNs）作为骨干网络[12, 13, 30, 45, 53]。随后，视觉Transformer（ViTs）[15]因其卓越性能而广受关注[5, 7, 10, 28, 29]。近期，选择性状态空间模型（Mamba）[17]被探索应用，展现出作为图像修复任务替代骨干网络的巨大潜力[18, 36]。

\*等额贡献 † 通讯作者

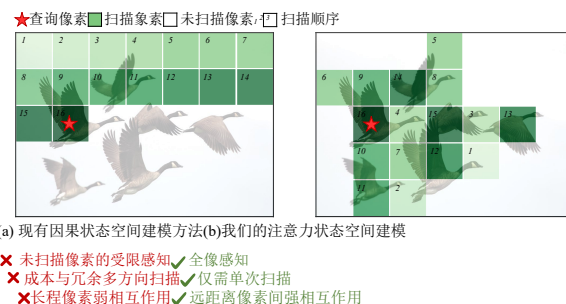


图1. (a) 现有方法[18]存在Mamba因果结构带来的负面影响（为提高展示清晰度未显示多方向扫描结果）。(b) 提出的MambaIRv2能够实现注意力状态空间建模，将类似ViT的非因果特性嵌入Mamba模型中。

尽管潜力巨大，现有的基于Mamba的方法仍面临重大挑战，尤其是由于其依赖于因果状态空间建模。具体而言，现有方法[18]通过预定义扫描规则对二维图像进行解构以生成一维标记序列。然而在Mamba中，每个像素仅基于扫描序列中其前序像素进行建模，即因果属性，这导致非因果图像修复任务中出现若干不利影响。首先，如图1(a)所示，查询像素仅能捕获前序像素信息而无法感知后续像素，导致图像中有效像素利用率不足。其次，固有的因果属性使得多方向扫描成为必要手段，现有方法[18, 36, 39]普遍采用该策略以缓解信息丢失问题。但这种多扫描方式不可避免地增加计算复杂度，尤其对高分辨率输入而言。此外，第3.3表明这些多方向扫描之间还存在大量信息冗余。第三，我们在第3节的研究结果表明，Mamba[17]在标记交互中容易出现长程衰减现象，即序列中距离较远的标记间交互作用减弱。因此，即使先前扫描过的距离较远但相关的像素也无法被查询像素有效利用。

In this work, we propose MambaIRv2, aiming to address the adverse effect of causal state-space modeling. Since the ViTs [15, 37] naturally support non-causal processing, our key idea is to integrate ViT-like non-causal modeling into the Mamba-based methods. To this end, we begin by delving deep into the connection between attention and state space for valuable insights. Our in-depth analysis in Sec. 4.1 reveals that the output matrix of the state-space equation resembles the query in the attention mechanism. This similarity inspires us to utilize the output matrix to “query” relevant pixels in the unscanned sequence. Benefiting from attending beyond the scanned sequences, this strategy also naturally eliminates the need for multi-directional scanning. Moreover, to encourage the interaction between distant but relevant pixels, we propose to restructure the image to place similar pixels spatially closer within the 1D sequence. In this way, it allows for semantic rather than spatial sequence modeling, thus mitigating the impact of long-range decay. Since the proposed method allows the Mamba to behave similarly to the attention, we thus refer to it as “**attentive state-space restoration**”.

Overall, we make three key contributions: **I.** We propose the **Attentive State-space Equation (ASE)**, which leverages the prompt learning [22] within the original state space equation of Mamba to query semantically similar pixels beyond the scanned sequences. In detail, the prompts are designed to represent sets of pixels that are similar across the entire image, and we then incorporate the representative prompts into the output matrix of the state-space equation through residual addition to derive our ASE. As the core component, the proposed ASE not only alleviates the causal nature of Mamba for improved performance but also enables single-pass scanning for boosted efficiency. **II.** We further develop the **Semantic Guided Neighboring (SGN)** to encourage strong interaction between distant yet similar pixels. Specifically, we first assign the corresponding semantic label to each pixel. Then we restructure the image based on these labels to generate the semantic-neighboring 1D sequence, where semantically similar pixels are also spatially close to each other. Thanks to the mitigation of the long-range decay of Mamba, SGN facilitates effective interaction between pixels that are distant in the original image. **III.** Integrating the two core modules and other auxiliary parts, we present MambaIRv2, an attentive state-space restoration method that equips Mamba’s state-space modeling with ViT-like non-causal capabilities. Extensive experiments demonstrate that MambaIRv2 significantly improves **both effectiveness and efficiency**. In particular, MambaIRv2 outperforms state-of-the-art Transformer-based baseline SRFormer [58] by 0.35dB on Urban100 dataset for  $2\times$  lightweight SR with 9.3% less parameters, and HAT [7] by 0.29dB for  $2\times$  classical SR on the Manga109 dataset.

## 2. Related Work

Recent years have witnessed great advancements in the domain of image restoration [2, 20, 23]. Early attempts usually adopt the convolutional neural networks (CNNs), such as SRCNN [13] for image super-resolution, DnCNN [45] for image denoising, and ARCNN [14] for JPEG compression artifacts reduction. To further enhance the performance of CNN-based methods, various techniques have been introduced. For instance, EDSR [25] employ the residual connection strategy to allow the training of very deep neural networks, RDN [53] uses the dense connection to improve model representation ability. RCAN [52] introduces the channel attention for selecting salient channels, followed by SAN [12] which uses the second-order attention for performance improvement. Despite the great progress of CNN-based methods, the convolution operator inherently restricts the receptive field to the local kernel, preventing interaction between distant pixels.

As Transformer [37] has proven its effectiveness in multiple computer vision tasks, applying Transformer for image restoration thus appears to be promising. However, the direct application of vanilla self-attention, which exhibits quadratic computational complexity with the input size, is costly and impractical. To improve the efficiency of attention, a variety of techniques have been developed. For example, IPT [5] divides one image into several small patches and processes each patch independently with self-attention. After that, SwinIR [29] further introduces the shifted window self-attention [32] to improve the performance. ART [44] and OminiSR [38] utilize sparse attention to expand the receptive field by enlarging the attention windows. GRL [28] adopts the anchor attention to learn the local, regional, and global image hierarchies. Recently, the ATD [49] uses the adaptive token dictionary to store input-agnostic knowledge, thus allowing the attention to attend information out of the local window.

To balance the efficient computation and global receptive fields, the Mamba [17] has recently been explored in image restoration with promising results. MambaIR [18] is among the first to introduce Mamba for image restoration and addresses two specific challenges, *i.e.*, local pixel forgetting and channel redundancy. Since then, the Mamba model has been explored in various image restoration tasks. FreqMamba [56] uses the state space model in the Fourier domain for image deraining to perceive global degradation. MambaLLIE [39] improves the state space equation to allow the locality enhancement for low-light image enhancement tasks. Moreover, Mamba has also achieved promising results in image dehazing [57], deblurring [16], and other tasks [3, 31, 34, 35, 40, 41, 50]. However, existing methods still struggle with the causal modeling nature of Mamba. Given image restoration as a non-causal task, this mismatch can lead to limited performance as well as inefficiency.



在本研究中，我们提出MambaIRv2模型，旨在解决因果状态空间建模带来的负面影响。由于视觉变分变换器（ViTs）[15, 37]天然支持非因果处理，我们的核心思路是将类ViT非因果建模方法整合到基于Mamba的框架中。为此，我们首先深入探究注意力机制与状态空间之间的关联以获取关键洞见。第4.1节的深度分析表明，状态空间方程的输出矩阵与注意力机制中的查询向量具有高度相似性。这种相似性启发我们利用输出矩阵对未扫描序列中的相关像素进行“查询”。通过利用超出扫描序列的注意力机制，该策略自然消除了多方向扫描的需求。此外，为促进远距离相关像素间的交互作用，我们提出对图像进行重构，使相似像素在1D序列中空间位置更趋接近。这种设计实现了语义而非空间序列建模，从而有效缓解长程衰减的影响。由于所提出的方法使Mamba能够表现出与注意力机制相似的行为，因此我们将该方法称为“**注意力状态空间恢复**”。

总体而言，我们做出了三项关键贡献：**I.**我们提出**注意力状态空间方程（ASE）**，该方法利用Mamba原始状态空间方程中的提示学习[22]来查询扫描序列之外语义相似的像素。具体而言，提示被设计为表征整个图像中相似像素的集合，随后通过残差加法将这些代表性提示整合到状态空间方程的输出矩阵中以推导出ASE。作为核心组件，所提出的ASE不仅缓解了Mamba的因果特性以提升性能，还实现了单遍扫描以提高效率。**II.**我们进一步开发了**语义引导邻近（SGN）**机制，以促进远距离但相似像素间的强交互作用。具体而言，我们首先为每个像素分配对应的语义标签，然后基于这些标签重构图像以生成语义邻近一维序列，其中语义相似像素在空间上也彼此接近。得益于对Mamba长程衰减的缓解，SGN能够促进原始图像中相距较远像素之间的有效交互。**III.**通过整合两个核心模块及其他辅助组件，我们提出MambaIRv2——一种具有ViT类非因果能力的注意力状态空间恢复方法，为Mamba的状态空间建模赋予了此类能力。大量实验表明，MambaIRv2显著提升了**效果与效率**。具体而言，在Urban100数据集上进行 $2\times$ 轻量级超分辨率时，MambaIRv2以9.3%更少的参数量较最先进的Transformer基线SRFormer[58]提升了0.35dB；在Manga109数据集上进行 $2\times$ 经典超分辨率时，较HAT[7]提升了0.29dB。

## 2. 相关工作

近年来，图像复原领域取得了重大进展[2, 20, 23]。早期尝试通常采用卷积神经网络（CNNs），例如用于图像超分辨率的SRCNN[13]、用于图像去噪的DnCNN[45]，以及用于JPEG压缩伪影消除的ARCNN[14]。为提升基于CNN方法的性能，多种技术被引入。例如，EDSR[25]采用残差连接策略以支持极深神经网络的训练，RDN[53]利用密集连接增强模型表征能力，RCAN[52]引入通道注意力机制筛选显著通道，随后SAN[12]通过二阶注意力机制提升性能。尽管基于CNN的方法取得显著进展，卷积算子固有的感受野局限性仍使像素间远距离交互无法实现。

由于Transformer[37]已在多项计算机视觉任务中证明其有效性，将其应用于图像修复领域显得颇具前景。然而，直接应用计算复杂度随输入尺寸呈二次增长的原始自注意力机制成本高昂且不切实际。为提升注意力效率，学界已开发出多种技术：例如IPT[5]将单幅图像分割为多个小块并独立处理每个块的自注意力机制；随后SwinIR[29]引入位移窗口自注意力[32]以提升性能；ART[44]与OminiSR[38]通过扩展注意力窗口利用稀疏注意力扩大感受野；GRL[28]采用锚点注意力学习局部、区域及全局图像层级结构；近期ATD[49]通过自适应标记词典存储输入无关知识，使注意力机制能够关注局部窗口之外的信息。

为平衡高效计算与全局感受野，Mamba[17]近期在图像修复领域得到探索并取得显著成果。MambaIR[18]是最早将Mamba应用于图像修复的模型之一，成功解决了局部像素遗忘与通道冗余两大挑战<sup>1</sup>。此后，Mamba模型在各类图像修复任务中持续被研究：FreqMamba[56]通过傅里叶域状态空间模型实现图像去噪以感知全局退化；MambaLLIE[39]改进状态空间方程以支持弱光图像增强任务中的局部增强效果。此外，Mamba在图像去雾[57]、去噪[16]等任务中也取得优异表现[3, 31, 34, 35, 40, 41, 50]。然而现有方法仍难以应对Mamba的因果建模特性。鉴于图像修复属于非因果任务，这种不匹配可能导致性能受限与效率低下。

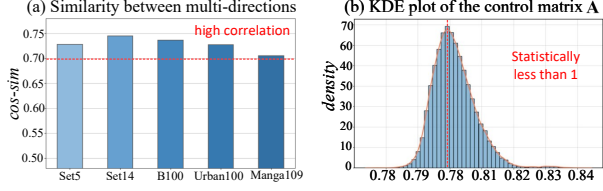


Figure 2. (a) We compute the cosine similarity of scanned features across all 4 directions and all layers in MambaIR [18]. (b) The kernel density estimation of the distribution of the control matrix in MambaIR [18].

### 3. Motivation

**Mamba-based Image Restoration.** The existing state-space restoration methods are mainly developed from the Mamba [17] architecture. Formally, the Mamba adopts the discrete state space equation to model the interaction among tokens:

$$\begin{aligned} h_i &= \bar{\mathbf{A}}h_{i-1} + \bar{\mathbf{B}}x_i, \\ y_i &= \mathbf{C}h_i + \mathbf{D}x_i, \end{aligned} \quad (1)$$

where the  $\bar{\mathbf{A}} = \exp(\Delta\mathbf{A})$  is the control matrix, the  $\bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I})\Delta\mathbf{B} \approx \Delta\mathbf{B}$  is the input matrix, and the  $\mathbf{C}$  is the output matrix. Eq. (1) indicates that the  $i$ -th token completely depends on its previous  $i - 1$  tokens, *i.e.*, the state-space modeling possesses causal properties. Although this causal nature is helpful for autoregressive tasks like NLP, it poses challenges for image restoration.

**Challenges from Causal Modeling.** Existing Mamba-based methods usually adopt a specific scanning strategy to unfold the 2D image into a 1D sequence for sequential modeling with Mamba. However, the  $i$ -th pixel can only see constrained  $i - 1$  pixels of the entire image, failing to globally utilize similar pixels. To this end, current methods typically employ multi-directional scans to allow for a broader receptive field, which is inevitably accompanied by an increase in computational complexity. Furthermore, as shown in Fig. 2(a), the similarity of different scanned sequences on all testing datasets reaches even above 0.7, indicating a high correlation with large redundancy. Moreover, the Mamba [17] itself also possesses long-range decay defects due to its causal nature. Specifically, the interaction between pixels can be quantitatively denoted by the power of control matrix  $\bar{\mathbf{A}}^k$ , where  $k$  is the pairwise distance (the proof is given in the *Suppl.*). In Fig. 2(b), we show that the learned  $\bar{\mathbf{A}}$  is statistically less than 1. As a result, the interaction  $\bar{\mathbf{A}}^k$  will become weak when two pixels are far apart, *i.e.*, a large  $k$ , indicating current methods fail to utilize distant but useful scanned pixels.

### 4. Attentive State Space Restoration

In the following, we aim to address the causal nature of Mamba through the proposed attentive state space restoration.

To get started, we visit the mathematical connection between state space and attention in Sec. 4.1 for insights of subsequent design. Then, we detail the specific techniques of our attentive state space module in Sec. 4.2. In Sec. 4.3, we give the overall architecture of the proposed methods.

#### 4.1. Bridging Attention and State-Space

As pointed out by [19], the state space has a strong relationship to attention, which may potentially offer insights to incorporate non-causal modeling ability into Mamba. In this section, we first reformulate attention and state space into the common form for comparison, followed by a detailed connection analysis.

**Reformulation of Attention.** Since the Mamba belongs to causal models with linear complexity, we adopt the corresponding causal linear attention [24] as its counterpart. Specifically, given the query, key and value matrix  $\mathbf{Q}$ ,  $\mathbf{K}$ ,  $\mathbf{V}$ , the output of linear attention is computed as follows:

$$y_i = \sum_{j=1}^i \frac{\mathbf{Q}_i \mathbf{K}_j^\top}{\sum_{t=1}^i \mathbf{Q}_i \mathbf{K}_t^\top} \mathbf{V}_j = \frac{\mathbf{Q}_i \left( \sum_{j=1}^i \mathbf{K}_j^\top \mathbf{V}_j \right)}{\mathbf{Q}_i \left( \sum_{t=1}^i \mathbf{K}_t^\top \right)}. \quad (2)$$

Denote the  $\mathbf{S}_i = \sum_{j=1}^i \mathbf{K}_j^\top \mathbf{V}_j$ ,  $\mathbf{Z}_i = \sum_{t=1}^i \mathbf{K}_t^\top$ , then the formulation of the linear attention can be rewritten as:

$$y_i = \mathbf{Q}_i \mathbf{S}_i / \mathbf{Q}_i \mathbf{Z}_i, \quad (3)$$

where  $\mathbf{S}_i = \mathbf{S}_{i-1} + \mathbf{K}_i^\top \mathbf{V}_i$ , and  $\mathbf{Z}_i = \mathbf{Z}_{i-1} + \mathbf{K}_i^\top$ . To allow subsequent connection analysis, we further reformulate Eq. (3) to the common form as follows:

$$\begin{aligned} \mathbf{S}_i &= \mathbf{I} \mathbf{S}_{i-1} + \mathbf{K}_i^\top \mathbf{V}_i, \\ y_i &= \mathbf{Q}_i \mathbf{S}_i / \mathbf{Q}_i \mathbf{Z}_i + \mathbf{O} x_i, \end{aligned} \quad (4)$$

where the  $\mathbf{I}$  and  $\mathbf{O}$  denotes identity and zero matrix, respectively.  $x_i$  is the input token at the  $i$ -th step.

**Reformulation of State Space.** Starting with state-space equation in Eq. (1), note that  $\bar{\mathbf{B}}x_i \approx \Delta\mathbf{B}x_i = \mathbf{B}(\Delta x_i)$ , Eq. (1) can then be reformulated to the common form as:

$$\begin{aligned} h_i &= \bar{\mathbf{A}}h_{i-1} + \mathbf{B}(\Delta x_i), \\ y_i &= \mathbf{C}h_i / \mathbf{I} + \mathbf{D}x_i. \end{aligned} \quad (5)$$

**Connection Analysis.** By comparing Eq. (4) and Eq. (5), it can be seen there is a close mathematical similarity between attention and state space, *i.e.*,  $h_i \sim \mathbf{S}_i$ ,  $\mathbf{B} \sim \mathbf{K}^\top$ , and  $\mathbf{C} \sim \mathbf{Q}$ . It should be noted that the  $\mathbf{Q}_i \mathbf{Z}_i$  in Eq. (4) is to ensure the attention score is normalized with summation 1 and can be approximately ignored if we relax the normalization constraints. Therefore, the above observation motivates us to delve deep into the output matrix  $\mathbf{C}$  in Eq. (1), which plays a role similar to the query in the attention mechanism. The core idea is to integrate the information of the unscanned sequence into  $\mathbf{C}$ , thus allowing  $\mathbf{C}$  to attentively “query” unseen pixels to facilitate the restoration of  $x_i$ .

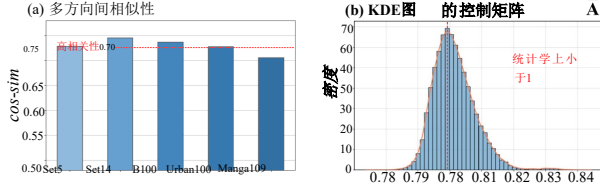


图2. (a) 我们计算了MambaIR中所有4个方向及所有层间扫描特征的余弦相似度[18]。 (b) MambaIR中控制矩阵分布的核密度估计[18]。

### 3. 动机

**基于Mamba的图像恢复。**现有的状态空间恢复方法主要基于Mamba[17]架构开发。从形式上讲，Mamba采用离散状态空间方程来建模标记之间的交互关系：

$$\begin{aligned} h_i &= \bar{\mathbf{A}}h_{i-1} + \bar{\mathbf{B}}x_i, \\ y_i &= \mathbf{C}h_i + \mathbf{D}x_i, \end{aligned} \quad (1)$$

其中 $\bar{\mathbf{A}} = \exp(\Delta\mathbf{A})$ 是控制矩阵， $\bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I})\Delta\mathbf{B} \approx \Delta\mathbf{B}$ 是输入矩阵， $\mathbf{C}$ 是输出矩阵。等式(1)表明第*i*个标记完全依赖于其前*i-1*个标记，即状态空间建模具有因果特性。尽管这种因果特性对自然语言处理等自回归任务有益，但对图像修复提出了挑战。**因果建模带来的挑战。**现有的基于Mamba的方法通常采用特定扫描策略将二维图像展开为一维序列以便与Mamba进行序列建模。然而，第*i*个像素只能观察到整个图像中受限的*i-1*个像素，无法全局利用相似像素。为此，现有方法通常采用多方向扫描技术以扩大感受野范围，但这必然会导致计算复杂度的提升。如图2(a)所示，所有测试数据集上不同扫描序列的相似度均超过0.7，表明其具有高度相关性且存在显著冗余性。此外，Mamba[17]模型本身因其因果特性也存在长程衰减缺陷。具体而言，像素间的相互作用可通过功率函数进行定量表征。

控制矩阵 $\bar{\mathbf{A}}$ ，其中 $\bar{a}_{ij}^k$ 表示成对距离（证明详见附录）。

图2(b)显示，学习得到的 $\bar{\mathbf{A}}$ 在统计上小于1。因此，互当两个像素间距较大时，动作 $\bar{\mathbf{A}}$ 会变得较弱，即， $k$ 值较大，这表明当前方法未能利用远处但有用的扫描像素。

### 4. 注意力型状态空间恢复

下文旨在通过提出的注意力状态空间恢复机制，探究Mamba的因果特性。

首先，我们在第4.1节探讨状态空间与注意力机制之间的数学联系，为后续设计提供理论依据。接着在第4.2节详细阐述注意力状态空间模块的具体实现技术。最后在第4.3节给出所提方法的整体架构设计。

#### 4.1. 注意力与状态空间的桥梁作用

如[19]所述，状态空间与注意力机制存在密切关联，这可能为将非因果建模能力整合至Mamba模型提供理论依据。本节首先将注意力机制与状态空间统一转化为通用表达形式以便对比分析，随后展开详细关联性研究。

**注意力机制的重构。**由于Mamba属于线性复杂度的因果模型，我们采用相应的因果线性注意力机制[24]作为其对应方案。具体而言，给定查询向量、键向量和值向量矩阵 $\mathbf{Q}$ 、 $\mathbf{K}$ 、 $\mathbf{V}$ ，线性注意力机制的输出计算如下：

$$y_i = \sum_{j=1}^i \frac{\mathbf{Q}_i \mathbf{K}_j^\top}{\sum_{t=1}^i \mathbf{Q}_i \mathbf{K}_t^\top} \mathbf{V}_j = \frac{\mathbf{Q}_i \left( \sum_{j=1}^i \mathbf{K}_j^\top \mathbf{V}_j \right)}{\mathbf{Q}_i \left( \sum_{t=1}^i \mathbf{K}_t^\top \right)}. \quad (2)$$

记 $\mathbf{S}_i = \sum_{j=1}^i \mathbf{K}_j^\top \mathbf{V}_j$ ， $\mathbf{Z}_i = \sum_{t=1}^i \mathbf{K}_t^\top$ ，此时线性注意力机制的公式可重写为：

$$y_i = \mathbf{Q}_i \mathbf{S}_i / \mathbf{Q}_i \mathbf{Z}_i, \quad (3)$$

其中 $\mathbf{S}_i = \mathbf{S}_{i-1} + \mathbf{K}_i^\top \mathbf{V}_i$ ，且 $\mathbf{Z}_i = \mathbf{Z}_{i-1} + \mathbf{K}_i^\top$ 。为便于后续连接分析，我们将等式(3)进一步改写为以下通用形式：

$$\begin{aligned} \mathbf{S}_i &= \mathbf{I} \mathbf{S}_{i-1} + \mathbf{K}_i^\top \mathbf{V}_i, \\ y_i &= \mathbf{Q}_i \mathbf{S}_i / \mathbf{Q}_i \mathbf{Z}_i + \mathbf{O} x_i, \end{aligned} \quad (4)$$

其中 $\mathbf{I}$ 和 $\mathbf{O}$ 分别表示单位矩阵和零矩阵。 $x_i$ 是第*i*步的输入标记。

**状态空间的重新表述。**从等式(1)中的状态空间方程出发，注意到 $\mathbf{B}x_i \approx \Delta \mathbf{B}x_i = \mathbf{B}(\Delta x_i)$ ，因此等式(1)可重新表述为以下通用形式：

$$\begin{aligned} h_i &= \mathbf{A}h_{i-1} + \mathbf{B}(\Delta x_i), \\ y_i &= \mathbf{C}h_i / \mathbf{I} + \mathbf{D}x_i. \end{aligned} \quad (5)$$

**连接分析。**通过比较等式(4)和等式(5)，可以看出注意力机制与状态空间之间存在密切的数学相似性，即， $h_i \sim \mathbf{S}_i$ 、 $\mathbf{B} \sim \mathbf{K}^\top$ 以及 $\mathbf{C} \sim \mathbf{Q}$ 。需要注意的是，等式(4)中的 $\mathbf{Q}_i \mathbf{Z}_i$ 是为了确保注意力分数通过求和1进行归一化，若放宽归一化约束则可近似忽略。因此，上述观察促使我们深入研究等式(1)中的输出矩阵 $\mathbf{C}$ ，其作用类似于注意力机制中的查询。核心思想是将未扫描序列的信息整合到 $\mathbf{C}$ 中，从而使 $\mathbf{C}$ 能够主动“查询”未见像素以促进 $x_i$ 的恢复。

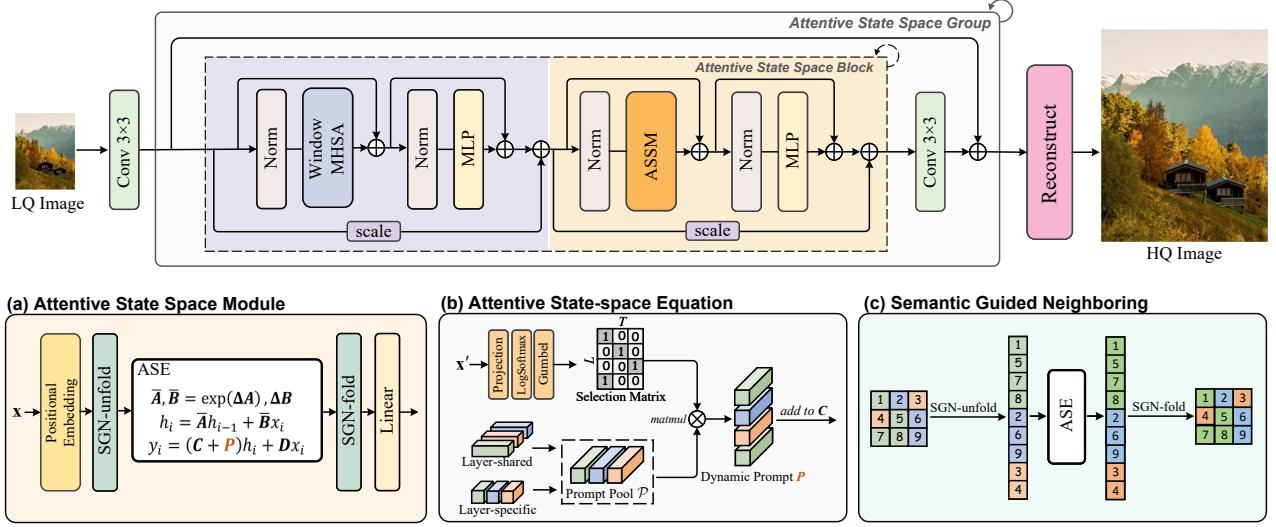


Figure 3. The overall architecture of our proposed MambaIRv2, as well as the (a) Attentive State Space Module (ASSM), (b) Attentive State-space Equation (ASE), and (c) Semantic Guided Neighboring (SGN).

## 4.2. Attentive State Space Module

In this section, we introduce the attentive state-space module (ASSM), which acts as the core block of our MambaIRv2 to enable non-causal modeling with Mamba. As shown in Fig. 3(a), given the input feature  $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ , where  $H$  and  $W$  are the height and width, respectively, and  $C$  is the channel dimension, we first apply the positional encoding [11] on  $\mathbf{x}$  to preserve the original structure information. After that, we propose the Semantic Guided Neighboring (SGN) to unfold the 2D image into 1D sequences for subsequent Attentive State-space Equation (ASE) modeling. Finally, another SGN is employed as the inverse operator of the previous one to fold the sequence back to the image followed by a linear projection to obtain the block output. More details are given below.

**Attentive State-space Equation.** As analyzed in Sec. 4.1, we aim to modify the output matrix  $\mathbf{C} \in \mathbb{R}^{L \times d}$ , where  $L = HW$  is the flattened image sequence length and  $d$  is the number of hidden states in Mamba, to globally query related pixels across the image. To this end, we propose the Attentive State-space Equation (ASE) which develops from the original state-space equation of Mamba but possesses a non-causal nature. As shown in Fig. 3(b), the proposed ASE incorporates prompts, which learn to represent a certain set of pixels with similar semantics, into  $\mathbf{C}$  to supplement the missing information of the unseen pixels. Specifically, we first construct the prompt pool  $\mathcal{P} \in \mathbb{R}^{T \times d}$ , where  $T$  is the number of prompts in  $\mathcal{P}$ . For parameterization of  $\mathcal{P}$ , we employ the semantic decoupling for better interpretability:

$$\mathcal{P} = \mathbf{M}\mathbf{N}, \quad \mathbf{M} \in \mathbb{R}^{T \times r}, \quad \mathbf{N} \in \mathbb{R}^{r \times d}, \quad (6)$$

where  $\mathbf{N}$  is shared across different blocks,  $\mathbf{M}$  is block-

specific, and  $r$  is the inner rank with  $r \ll \min\{T, d\}$ . The main idea behind the semantic decoupling is that we want different blocks to share similar feature space, *i.e.*,  $\mathbf{N}$  is shared, while the combination coefficients of the shared features can be varying for different blocks *i.e.*,  $\mathbf{M}$  is specific.

After that, we develop routing strategies to select from  $\mathcal{P}$  to obtain  $L$  instance-specific prompts  $\mathbf{P} \in \mathbb{R}^{L \times d}$ , which will be added into  $\mathbf{C}$  to include information of unscanned pixels. In detail, given the flattened input feature  $\mathbf{x}' \in \mathbb{R}^{L \times C}$ , we employ a linear layer to project the channel dimension of  $\mathbf{x}'$  from  $C$  to  $T$ , followed by the LogSoftmax to predict the log probability, which indicates the probability of each prompt in  $\mathcal{P}$  being sampled by  $\mathbf{x}'_i, i = 1, 2, \dots, L$ . After that, we introduce the gumbel-softmax [21] trick on the log probability to allow differentiable prompt selection to obtain the one-hot routing matrix  $\mathbf{R} \in \mathbb{R}^{L \times T}$ . Then, the instance-specific prompt  $\mathbf{P} \in \mathbb{R}^{L \times d}$  is generated through the matrix multiplication as  $\mathbf{P} = \mathbf{R}\mathcal{P}$ . Finally, we incorporate  $\mathbf{P}$  into  $\mathbf{C}$  through residual addition to formulate our attentive state-space equation:

$$\begin{aligned} h_i &= \bar{\mathbf{A}}h_{i-1} + \bar{\mathbf{B}}x_i, \\ y_i &= (\mathbf{C} + \mathbf{P})h_i + \mathbf{D}x_i. \end{aligned} \quad (7)$$

The learned prompts allow for the attention-like capability to query pixels across the whole image, and we also present a visualization on the attentive map in Fig. 5. By injecting the prompts that represent the set of similar pixels, the proposed ASE can effectively alleviate constrained perception for unscanned pixels. As another advantage, it allows scanning with only one single direction, eliminating the high computational cost and redundancy of multi-directional scans in existing methods.



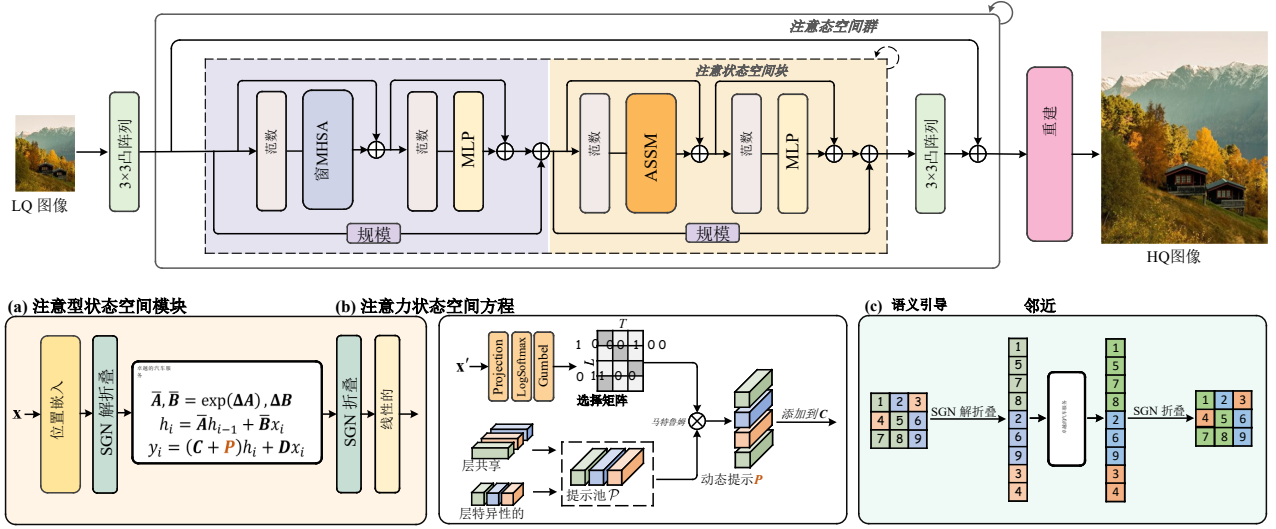


图3.我们提出的MambaIRv2整体架构，包括(a)注意力状态空间模块(ASSM)、(b)注意力状态空间方程(ASE)以及(c)语义引导邻近机制(SGN)。

#### 4.2. 注意型状态空间模块

在本节中，我们介绍注意力状态空间模块（ASSM），该模块作为MambaIRv2的核心组件，用于实现Mamba的非因果建模。如图3(a)所示，给定输入特征  $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ （其中  $H$  和  $W$  分别为图像高度和宽度， $C$  为通道维度），我们首先对  $\mathbf{x}$  应用位置编码[11]以保留原始结构信息。随后，我们提出语义引导邻域（SGN）将二维图像展开为一维序列，以便后续进行注意力状态空间方程（ASE）建模。最后，采用另一个 SGN 作为前者的逆运算符将序列折叠回图像，并通过线性投影获得模块输出。更多细节如下所述。

**注意力状态空间方程。**如第4.1节所述，我们旨在修改输出矩阵  $\mathbf{C} \in \mathbb{R}^{L \times d}$ （其中  $L=HW$  表示图像序列展平长度， $d$  为Mamba模型中的隐藏状态数量），以全局查询图像中相关像素。为此，我们提出注意力状态空间方程（ASE），该方程基于Mamba原始状态空间方程发展而来，但具有非因果特性。如图3(b)所示，所提出的ASE将提示词融入  $\mathbf{C}$  中，这些提示词通过学习表征具有相似语义的像素集合来补充未见像素的缺失信息。具体而言，我们首先构建提示池  $\mathbf{P} \in \mathbb{R}^{T \times d}$ （其中  $T$  为  $\mathbf{P}$  中提示词数量）。为提升可解释性， $\mathbf{P}$  的参数化采用语义解耦方法：

$$\mathbf{P} = \mathbf{M}\mathbf{N}, \quad \mathbf{M} \in \mathbb{R}^{T \times r}, \quad \mathbf{N} \in \mathbb{R}^{r \times d}, \quad (6)$$

其中  $\mathbf{N}$  在不同区块间共享， $\mathbf{M}$  为区块-

具体而言， $r$  是内秩，且  $r = \min\{T, d\}$ 。语义解耦的核心思想在于：我们希望不同模块共享相似的特征空间，即共享  $\mathbf{N}$ ，而共享特征的组合系数可因模块不同而变化，即  $\mathbf{M}$  具有模块特异性。

之后，我们开发路由策略从  $\mathbf{P}$  中进行选择以获得  $L$  个实例特定提示  $\mathbf{P} \in \mathbb{R}^{L \times d}$ ，这些提示将被添加到  $\mathbf{C}$  中以包含未扫描像素的信息。具体而言，给定展平后的输入特征  $\mathbf{x}' \in \mathbb{R}^{L \times C}$ ，我们采用线性层将  $\mathbf{x}'$  的通道维度从  $C$  投影到  $T$ ，随后通过  $\text{LogSoftmax}$  预测对数概率，该概率表示  $\mathbf{P}$  中每个提示被  $\mathbf{x}'_i, i=1, 2, \dots, L$  采样的概率。接着，我们在对数概率上引入  $\text{gumbel-softmax}$ [21] 技巧以实现可微分提示选择，从而获得独热路由矩阵  $\mathbf{R} \in \mathbb{R}^{L \times T}$ 。随后，通过矩阵乘法生成实例特定提示  $\mathbf{P} \in \mathbb{R}^{L \times d}$ ，即  $\mathbf{P} = \mathbf{R}\mathbf{P}$ 。最后，我们通过残差添加将  $\mathbf{P}$  融入  $\mathbf{C}$  中，从而构建注意力状态空间方程：

$$\begin{aligned} h_i &= \bar{\mathbf{A}}h_{i-1} + \bar{\mathbf{B}}x_i, \\ y_i &= (\mathbf{C} + \mathbf{P})h_i + \mathbf{D}x_i. \end{aligned} \quad (7)$$

经过学习的提示词能够实现类似注意力机制的功能，从而实现对整幅图像像素的查询。我们在图5中展示了注意力图的可视化效果。通过注入代表相似像素集合的提示词，所提出的ASE方法能有效缓解未扫描像素的感知限制。另一个优势在于，该方法仅需单向扫描即可完成检测，从而避免了现有方法中多方向扫描带来的高计算成本和冗余问题。



**Semantic Guided Neighboring.** As pointed out in Sec. 3, the causal modeling property of Mamba leads to the detrimental effects of long-range decay. In existing Mamba-based image restoration methods, pixels that are distant in the original image are usually still far apart in the unfolded sequence, causing the weak utilization of the query pixel for the already scanned pixels which are spatially distant but similar. To this end, we propose the Sematic Guided Neighboring (SGN) as shown in Fig. 3(c). Our key insight is that different from the autoregressive language modeling, the image restoration is a non-causal task and all pixels are observable at once, therefore we can re-define the token neighborhood to enable semantically similar tokens to be spatially closer in the unfolded sequence. Following this idea, we first determine the semantic label of each pixel. Note that the routing matrix  $\mathbf{R}$  in the ASE, which has learned the prompt category of each pixel, we thus employ this off-the-shelf semantics to restructure the image. Specifically, we propose the SGN-unfold which groups pixels with the  $i$ -th prompt category together to form the  $i$ -th semantic group and then combines different groups according to the category value  $i$  to generate the semantic-neighbored sequence. After that, we feed this sequence into the proposed ASE for state-space modeling. At last, we employ the SGN-fold which performs as the inverse transformation of SGN-unfold to reshape the semantic-space sequence back to the spatial-space feature map to obtain the output.

### 4.3. Overall Network Architecture

The proposed ASSM can efficiently capture the global dependencies using the Mamba model. We then further consider modeling local interactions, which have been shown crucial for Mamba-based approaches [18, 39]. Since the scanning only once in the ASSM provides more parameter budgets, we thus opt for the powerful window multi-head self-attention (MHSA) [32] to enhance local interactions within the window, which together with the ASSM compose the basic elements of our MambaIRv2. As shown in Fig. 3, given a low-quality image as the input, we first utilize the  $3 \times 3$  convolution layer to extract shallow features. Then the shallow features are fed into several Attentive State Space Groups (ASSGs), where each groups contain multiple Attentive State Space Blocks (ASSBs). For each ASSB, we consider progressive local-to-global modeling to form the image hierarchy [28]. We use Norm and Token Mixer, followed by Norm and FFN to form the template, and employ window MHSA and ASSM as the instantiations for the Token Mixer of the local and global parts, respectively. In addition, two residual connections with learnable scales are introduced [9, 18]. After the ASSGs, we utilize the task-specific reconstruction modules, *e.g.*, pixelshuffle for super-resolution, and convolution for denoising, to obtain the high-quality image output.

Table 1. Ablation on the effectiveness of different components.

| MHSA | ASE | SGN | Urban100     |               | Manga109     |               |
|------|-----|-----|--------------|---------------|--------------|---------------|
|      |     |     | PSNR         | SSIM          | PSNR         | SSIM          |
| ✓    |     |     | 32.89        | 0.9343        | 39.11        | 0.9772        |
| ✓    | ✓   |     | 32.94        | 0.9351        | 39.20        | 0.9780        |
| ✓    | ✓   | ✓   | <b>32.97</b> | <b>0.9355</b> | <b>39.24</b> | <b>0.9784</b> |

Table 2. Ablation experiments on different injection positions of the learnable prompts in the ASE.

| positions | Set14        |               | Urban100     |               | Manga109     |               |
|-----------|--------------|---------------|--------------|---------------|--------------|---------------|
|           | PSNR         | SSIM          | PSNR         | SSIM          | PSNR         | SSIM          |
| B         | 33.97        | 0.9215        | 32.96        | <b>0.9356</b> | 39.23        | 0.9781        |
| $\Delta$  | 33.92        | 0.9211        | 32.93        | 0.9350        | 39.19        | 0.9779        |
| y         | 33.97        | 0.9210        | 32.94        | 0.9351        | 39.21        | 0.9782        |
| C         | <b>33.99</b> | <b>0.9216</b> | <b>32.97</b> | 0.9355        | <b>39.24</b> | <b>0.9784</b> |

## 5. Experiments

Following previous image restoration works [18, 29], we conduct experiments on three representative image restoration tasks, *i.e.*, image super-resolution including classic SR and lightweight SR, JPEG compression artifact reduction (JPEG CAR), and Gaussian color image denoising.

### 5.1. Experimental Settings

In accordance with previous works, we perform data augmentation by applying horizontal flips and random rotations of  $90^\circ$ ,  $180^\circ$ , and  $270^\circ$ . Additionally, we crop the original images into  $64 \times 64$  patches for image SR and  $128 \times 128$  patches for image denoising during training. For image SR, we use the pre-trained weights from the  $2\times$  model to initialize those of  $3\times$  and  $4\times$  and halve the learning rate and total training iterations to reduce training time [30]. To ensure a fair comparison, we adjust the training batch size to 32 for image SR and 8 for image denoising and JPEG CAR. We employ the Adam [26] as the optimizer for training our MambaIRv2 with  $\beta_1 = 0.9$ ,  $\beta_2 = 0.999$ . Similar to previous training protocol [29], we use the  $L_1$  loss for image SR, and Charbonnier loss [4] for denoising and JPEG CAR. The initial learning rate is set at  $2 \times 10^{-4}$  and is halved when the training iteration reaches specific milestones. For classic image SR, we provide three variants with different parameters including the small, base, and large versions (MambaIRv2-S, MambaIRv2-B, MambaIRv2-L). Due to the page limit, more details are provided in the *Suppl.*

### 5.2. Ablation Study

We conduct ablations with MambaIRv2-light  $2\times$  SR model trained for 250K iterations on the DIV2K dataset.

**Effectiveness of Different Components.** As the core module of MambaIRv2, the Attentive State Space Module (ASSM), which contains the Attentive State-space Equation (ASE) and the Semantic Guided Neighboring (SGN), plays an important role in Mamba-based global modeling. In this ablation, we design three settings to verify the role of the different components. In the first setup, we directly re-

**语义引导邻域。**如第3节所述，Mamba的因果建模特性会导致长程衰减的负面影响。在现有的基于Mamba的图像修复方法中，原始图像中相距较远的像素在展开序列中通常仍保持较大间距，这使得已扫描像素在空间上虽相距较远但具有相似性时，查询像素的利用率较低。为此，我们提出如图3(c)所示的语义引导邻域（SGN）。我们的核心洞见在于：与自回归语言模型不同，图像修复属于非因果任务且所有像素可同时观测，因此可通过重新定义标记邻域使语义相似标记在展开序列中空间位置更接近。基于此思路，我们首先确定每个像素的语义标签。需注意，ASE中的路由矩阵 $\mathbf{R}$ 已学习到各像素的提示类别，因此我们采用现成语义信息对图像进行重构。具体而言，我们提出SGN展开算法：首先将属于第 $i$ 个提示类别的像素进行分组，形成第 $i$ 个语义组；随后根据类别值 $i$ 对不同组别进行整合，生成语义邻近序列。接着将该序列输入所提出的ASE模型进行状态空间建模。最后采用SGN折叠算法——作为SGN展开的逆变换操作，将语义空间序列重构为空间特征图，最终获得输出结果。

#### 4.3. 总体网络架构

所提出的 ASSM 能够利用Mamba模型高效捕捉全局依赖关系。我们进一步考虑了局部交互建模问题，已有研究表明这对于基于Mamba的方法至关重要[18, 39]。由于在 ASSM 中仅进行一次扫描可提供更多参数预算，因此我们选择采用强大的窗口多头自注意力机制（MHSA）[32]来增强窗口内的局部交互，这些机制与ASSM共同构成了MambaIRv2的基础要素。如图3所示，当输入低质量图像时，我们首先通过 $3 \times 3$ 卷积层提取浅层特征，随后将这些特征输入多个注意力状态空间组（ASSGs），每个组内包含多个注意力状态空间块（ASSBs）。针对每个ASSB，我们采用渐进式局部到全局建模方法构建图像层级结构[28]。通过使用Norm和Token Mixer，再结合Norm和FFN生成模板，并分别采用窗口MHSA和ASSM作为局部与全局部分Token Mixer的具体实现方式。此外，我们引入了两个具有可学习尺度的残差连接[9, 18]。在ASSGs之后，我们利用任务特定的重建模块（例如用于超分辨率的pixelshuffle和用于去噪的卷积），以获得高质量的图像输出。

表1. 不同组件消融术的有效性比较

| MHSA | ASE | SGN | 城市100        |               | Manga109     |               |
|------|-----|-----|--------------|---------------|--------------|---------------|
|      |     |     | PSNR         | SSIM          | PSNR         | SSIM          |
| ✓    |     |     | 32.89        | 0.9343        | 39.11        | 0.9772        |
| ✓    | ✓   |     | 32.94        | 0.9351        | 39.20        | 0.9780        |
| ✓    | ✓   | ✓   | <b>32.97</b> | <b>0.9355</b> | <b>39.24</b> | <b>0.9784</b> |

表2. ASE中可学习提示不同注射位置的消融实验

| 职位       | Set14        |               | 城市100        |               | Manga109     |               |
|----------|--------------|---------------|--------------|---------------|--------------|---------------|
|          | PSNR         | SSIM          | PSNR         | SSIM          | PSNR         | SSIM          |
| B        | 33.97        | 0.9215        | 32.96        | <b>0.9356</b> | 39.23        | 0.9781        |
| $\Delta$ | 33.92        | 0.9211        | 32.93        | 0.9350        | 39.19        | 0.9779        |
| y        | 33.97        | 0.9210        | 32.94        | 0.9351        | 39.21        | 0.9782        |
| C        | <b>33.99</b> | <b>0.9216</b> | <b>32.97</b> | 0.9355        | <b>39.24</b> | <b>0.9784</b> |

## 5. 实验

基于先前的图像修复研究[18、29]，我们在三个代表性图像修复任务上开展实验，即：图像超分辨率（包括经典超分辨率和轻量级超分辨率）、JPEG压缩伪影消除（JPEG CAR）以及高斯彩色图像去噪。

### 5.1. 实验设置

与先前研究一致，我们通过应用水平翻转及 $90^\circ$ 、 $180^\circ$ 、 $270^\circ$ 随机旋转进行数据增强。此外，在训练过程中将原始图像裁剪为 $64 \times 64$ 像素块用于图像超分辨率， $128 \times 128$ 像素块用于图像去噪。对于图像超分辨率，我们采用 $2 \times$ 模型的预训练权重初始化 $3 \times$ 和 $4 \times$ 模型权重，并将学习率与总训练迭代次数减半以缩短训练时间[30]。为确保公平比较，我们将图像超分辨率的训练批次大小调整为32，图像去噪及JPEG CAR则调整为8。采用Adam[26]作为MambaIRv2模型的优化器，设置 $\zeta_1=0.9$ ， $\zeta_2=0.999$ 。与既往训练方案[29]类似，图像超分辨率采用 $L_1$ 损失函数，去噪及JPEG CAR采用Charbonnier损失[4]。初始学习率设为 $2 \times 10^{-4}$ ，并在训练迭代达到特定里程碑时减半。对于经典图像超分辨率（SR），我们提供了三种不同参数的变体，包括小尺寸、基础版和大尺寸版本（MambaIRv2-S、MambaIRv2-B、MambaIRv2-L）。由于篇幅限制，更多细节详见补充材料。

### 5.2. 消融研究

我们使用在DIV2K数据集上经过25万次迭代训练的MambaIRv2-light  $2 \times$  SR模型进行消融实验。**不同组件的有效性。**作为MambaIRv2的核心模块，包含注意力状态空间方程（ASE）和语义引导邻近机制（SGN）的注意力状态空间模块（ASSM），在基于Mamba的全局建模中发挥着关键作用。在本次消融实验中，我们设计了三种实验方案以验证不同组分的作用。在第一种方案中，我们直接重新

Table 3. Quantitative comparison on *lightweight image super-resolution* with state-of-the-art methods. The best and the second best results are in red and blue.

| Method              | scale | #param | MACs   | Set5  |        | Set14 |        | BSDS100 |        | Urban100 |        | Manga109 |        |
|---------------------|-------|--------|--------|-------|--------|-------|--------|---------|--------|----------|--------|----------|--------|
|                     |       |        |        | PSNR  | SSIM   | PSNR  | SSIM   | PSNR    | SSIM   | PSNR     | SSIM   | PSNR     | SSIM   |
| CARN [1]            | 2×    | 1,592K | 222.8G | 37.76 | 0.9590 | 33.52 | 0.9166 | 32.09   | 0.8978 | 31.92    | 0.9256 | 38.36    | 0.9765 |
| LatticeNet [33]     | 2×    | 756K   | 169.5G | 38.13 | 0.9610 | 33.78 | 0.9193 | 32.25   | 0.9005 | 32.43    | 0.9302 | -        | -      |
| SwinIR-light [29]   | 2×    | 910K   | 244.2G | 38.14 | 0.9611 | 33.86 | 0.9206 | 32.31   | 0.9012 | 32.76    | 0.9340 | 39.12    | 0.9783 |
| MambaIR-light [18]  | 2×    | 905K   | 334.2G | 38.13 | 0.9610 | 33.95 | 0.9208 | 32.31   | 0.9013 | 32.85    | 0.9349 | 39.20    | 0.9782 |
| ELAN [51]           | 2×    | 621K   | 203.1G | 38.17 | 0.9611 | 33.94 | 0.9207 | 32.30   | 0.9012 | 32.76    | 0.9340 | 39.11    | 0.9782 |
| SRFormer-light [58] | 2×    | 853K   | 236.3G | 38.23 | 0.9613 | 33.94 | 0.9209 | 32.36   | 0.9019 | 32.91    | 0.9353 | 39.28    | 0.9785 |
| MambaIRv2-light     | 2×    | 774K   | 286.3G | 38.26 | 0.9615 | 34.09 | 0.9221 | 32.36   | 0.9019 | 33.26    | 0.9378 | 39.35    | 0.9785 |
| CARN [1]            | 3×    | 1,592K | 118.8G | 34.29 | 0.9255 | 30.29 | 0.8407 | 29.06   | 0.8034 | 28.06    | 0.8493 | 33.50    | 0.9440 |
| LatticeNet [33]     | 3×    | 765K   | 76.3G  | 34.53 | 0.9281 | 30.39 | 0.8424 | 29.15   | 0.8059 | 28.33    | 0.8538 | -        | -      |
| SwinIR-light [29]   | 3×    | 918K   | 111.2G | 34.62 | 0.9289 | 30.54 | 0.8463 | 29.20   | 0.8082 | 28.66    | 0.8624 | 33.98    | 0.9478 |
| MambaIR-light       | 3×    | 913K   | 148.5G | 34.63 | 0.9288 | 30.54 | 0.8459 | 29.23   | 0.8084 | 28.70    | 0.8631 | 34.12    | 0.9479 |
| ELAN [51]           | 3×    | 629K   | 90.1G  | 34.61 | 0.9288 | 30.55 | 0.8463 | 29.21   | 0.8081 | 28.69    | 0.8624 | 34.00    | 0.9478 |
| SRformer-light [58] | 3×    | 861K   | 105.4G | 34.67 | 0.9296 | 30.57 | 0.8469 | 29.26   | 0.8099 | 28.81    | 0.8655 | 34.19    | 0.9489 |
| MambaIRv2-light     | 3×    | 781K   | 126.7G | 34.71 | 0.9298 | 30.68 | 0.8483 | 29.26   | 0.8098 | 29.01    | 0.8689 | 34.41    | 0.9497 |
| CARN [1]            | 4×    | 1,592K | 90.9G  | 32.13 | 0.8937 | 28.60 | 0.7806 | 27.58   | 0.7349 | 26.07    | 0.7837 | 30.47    | 0.9084 |
| LatticeNet [33]     | 4×    | 777K   | 43.6G  | 32.30 | 0.8962 | 28.68 | 0.7830 | 27.62   | 0.7367 | 26.25    | 0.7873 | -        | -      |
| SwinIR-light [29]   | 4×    | 930K   | 63.6G  | 32.44 | 0.8976 | 28.77 | 0.7858 | 27.69   | 0.7406 | 26.47    | 0.7980 | 30.92    | 0.9151 |
| MambaIR-light [18]  | 4×    | 924K   | 84.6G  | 32.42 | 0.8977 | 28.74 | 0.7847 | 27.68   | 0.7400 | 26.52    | 0.7983 | 30.94    | 0.9135 |
| ELAN [51]           | 4×    | 640K   | 54.1G  | 32.43 | 0.8975 | 28.78 | 0.7858 | 27.69   | 0.7406 | 26.54    | 0.7982 | 30.92    | 0.9150 |
| SRformer-light [58] | 4×    | 873K   | 62.8G  | 32.51 | 0.8988 | 28.82 | 0.7872 | 27.73   | 0.7422 | 26.67    | 0.8032 | 31.17    | 0.9165 |
| MambaIRv2-light     | 4×    | 790K   | 75.6G  | 32.51 | 0.8992 | 28.84 | 0.7878 | 27.75   | 0.7426 | 26.82    | 0.8079 | 31.24    | 0.9182 |

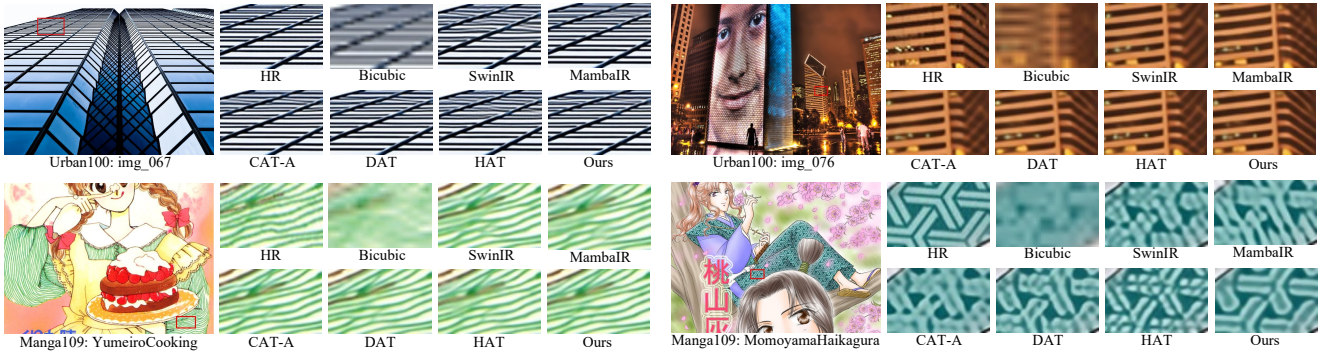


Figure 4. Qualitative comparison of our MambaIRv2 with different methods on 4× classic image SR.

move the ASSM, leading to a pure Attention variant. In the second setup, we add the proposed ASSM but remove the SGN. The third setup corresponds to our proposed method. As shown in Tab. 1, only using window attention limits the receptive field to the local window, limiting the performance. Further, the addition of the ASE, which allows for the query of similar pixels across images, improves the performance by 0.05 dB/0.09 dB in Urban100/Manga109. Finally, after introducing SGN which can effectively overcome the long-range decay of Mamba, the final model achieves the best performance of 32.97/39.24 dB PSNR on Urban100/Manga109. The experiments validate the effectiveness of different components in the proposed method.

**Ablation on Attentive State-space Equation.** In Sec. 4.1, we pointed out that the matrix  $C$  in the state-space equation in fact behaves similarly to the Query in the attention, following which we propose the ASE to insert global prompts into  $C$  by residual addition. In Tab. 2, we explore other inserting positions of the prompts in ASE. Since  $B$  can be

analogized to the Key in attention, inserting the prompts into  $B$  also yields decent results, but slightly inferior to adding to  $C$ . We attribute this to the fact that  $C$  is closer to the output end in the state-space equation Eq. (1), leading to a greater impact on the final result. Further, inserting prompts to the discrete time-step  $\Delta$  and the output  $y$  both fail to give satisfactory performance, which justifies our focus on the matrix  $C$  in the proposed ASE. Due to the page limit, we provide more ablation experiments in the *Suppl.*.

### 5.3. Comparison on Image Super-Resolution

**Lightweight Image Super-Resolution.** Following previous works [33, 58], we also report the number of parameters (#param) and MACs (upsampling a low-resolution image to  $1280 \times 720$  resolution) as the efficiency metric. The results in Tab. 3 show that our MambaIRv2 outperforms the state-of-the-art method SRFormer [58] using even significantly fewer parameters. For instance, our MambaIRv2-light outperforms SRFormer-light by 0.35dB on 2× Urban100 with



表3. 轻量级图像超分辨率与最先进方法的定量比较。最佳结果和次优结果分别用红色和蓝色表示。

| 方法                  | 规模 | #参数    | 多通道控制器 | Set5  |        | Set14 |        | BSDS100 |        | 城市100 |        | Manga109 |        |
|---------------------|----|--------|--------|-------|--------|-------|--------|---------|--------|-------|--------|----------|--------|
|                     |    |        |        | PSNR  | SSIM   | PSNR  | SSIM   | PSNR    | SSIM   | PSNR  | SSIM   | PSNR     | SSIM   |
| CARN [1]            | 2× | 1,592K | 222.8G | 37.76 | 0.9590 | 33.52 | 0.9166 | 32.09   | 0.8978 | 31.92 | 0.9256 | 38.36    | 0.9765 |
| LatticeNet [33]     | 2× | 756K   | 169.5G | 38.13 | 0.9610 | 33.78 | 0.9193 | 32.25   | 0.9005 | 32.43 | 0.9302 | -        | -      |
| SwinIR-light [29]   | 2× | 910K   | 244.2G | 38.14 | 0.9611 | 33.86 | 0.9206 | 32.31   | 0.9012 | 32.76 | 0.9340 | 39.12    | 0.9783 |
| MambaIR-light [18]  | 2× | 905K   | 334.2G | 38.13 | 0.9610 | 33.95 | 0.9208 | 32.31   | 0.9013 | 32.85 | 0.9349 | 39.20    | 0.9782 |
| ELAN [51]           | 2× | 621K   | 203.1G | 38.17 | 0.9611 | 33.94 | 0.9207 | 32.30   | 0.9012 | 32.76 | 0.9340 | 39.11    | 0.9782 |
| SR前轻型[58]           | 2× | 853K   | 236.3G | 38.23 | 0.9613 | 33.94 | 0.9209 | 32.36   | 0.9019 | 32.91 | 0.9353 | 39.28    | 0.9785 |
| MambaIRv2-light     | 2× | 774K   | 286.3G | 38.26 | 0.9615 | 34.09 | 0.9221 | 32.36   | 0.9019 | 33.26 | 0.9378 | 39.35    | 0.9785 |
| CARN [1]            | 3× | 1,592K | 118.8G | 34.29 | 0.9255 | 30.29 | 0.8407 | 29.06   | 0.8034 | 28.06 | 0.8493 | 33.50    | 0.9440 |
| LatticeNet [33]     | 3× | 765K   | 76.3G  | 34.53 | 0.9281 | 30.39 | 0.8424 | 29.15   | 0.8059 | 28.33 | 0.8538 | -        | -      |
| SwinIR-light [29]   | 3× | 918K   | 111.2G | 34.62 | 0.9289 | 30.54 | 0.8463 | 29.20   | 0.8082 | 28.66 | 0.8624 | 33.98    | 0.9478 |
| MambaIR-light       | 3× | 913K   | 148.5G | 34.63 | 0.9288 | 30.54 | 0.8459 | 29.23   | 0.8084 | 28.70 | 0.8631 | 34.12    | 0.9479 |
| ELAN [51]           | 3× | 629K   | 90.1G  | 34.61 | 0.9288 | 30.55 | 0.8463 | 29.21   | 0.8081 | 28.69 | 0.8624 | 34.00    | 0.9478 |
| SRformer-light [58] | 3× | 861K   | 105.4G | 34.67 | 0.9296 | 30.57 | 0.8469 | 29.26   | 0.8099 | 28.81 | 0.8655 | 34.19    | 0.9489 |
| MambaIRv2-light     | 3× | 781K   | 126.7G | 34.71 | 0.9298 | 30.68 | 0.8483 | 29.26   | 0.8098 | 29.01 | 0.8689 | 34.41    | 0.9497 |
| CARN [1]            | 4× | 1,592K | 90.9G  | 32.13 | 0.8937 | 28.60 | 0.7806 | 27.58   | 0.7349 | 26.07 | 0.7837 | 30.47    | 0.9084 |
| LatticeNet [33]     | 4× | 777K   | 43.6G  | 32.30 | 0.8962 | 28.68 | 0.7830 | 27.62   | 0.7367 | 26.25 | 0.7873 | -        | -      |
| SwinIR-light [29]   | 4× | 930K   | 63.6G  | 32.44 | 0.8976 | 28.77 | 0.7858 | 27.69   | 0.7406 | 26.47 | 0.7980 | 30.92    | 0.9151 |
| MambaIR-light [18]  | 4× | 924K   | 84.6G  | 32.42 | 0.8977 | 28.74 | 0.7847 | 27.68   | 0.7400 | 26.52 | 0.7983 | 30.94    | 0.9135 |
| ELAN [51]           | 4× | 640K   | 54.1G  | 32.43 | 0.8975 | 28.78 | 0.7858 | 27.69   | 0.7406 | 26.54 | 0.7982 | 30.92    | 0.9150 |
| SRformer-light [58] | 4× | 873K   | 62.8G  | 32.51 | 0.8988 | 28.82 | 0.7872 | 27.73   | 0.7422 | 26.67 | 0.8032 | 31.17    | 0.9165 |
| MambaIRv2-light     | 4× | 790K   | 75.6G  | 32.51 | 0.8992 | 28.84 | 0.7878 | 27.75   | 0.7426 | 26.82 | 0.8079 | 31.24    | 0.9182 |

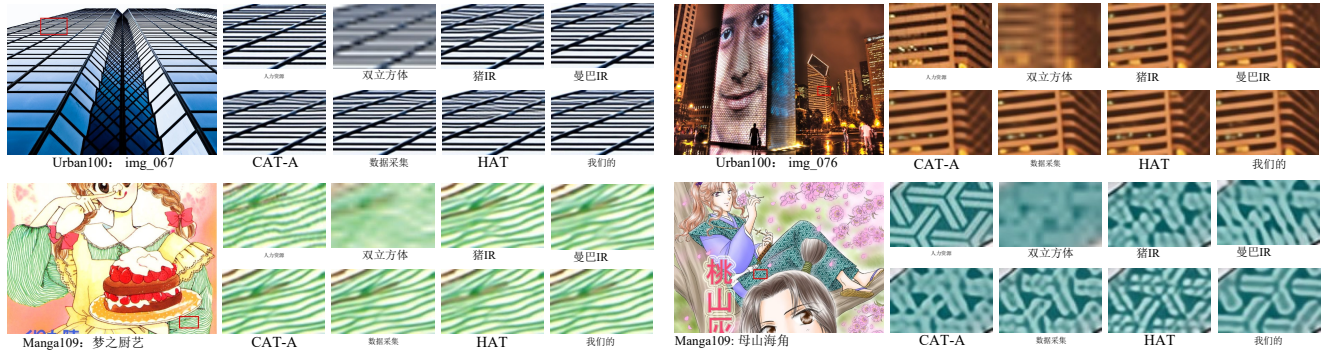


图4. 我们MambaIRv2与不同方法在4 ×经典图像超分辨率数据集上的定性比较。

移动 ASSM，从而得到纯注意力变体。在第二种设置中，我们添加了提出的 ASSM 但移除了 SGN。第三种设置对应于我们提出的方法。如表1所示，仅使用窗口注意力会将感受野限制在局部窗口内，导致性能受限。此外，通过引入允许跨图像查询相似像素的 ASE，我们在Urban100/Manga109数据集上将性能提升了0.05 dB/0.09 dB。最后，在引入能有效克服Mamba长程衰减的 SGN 后，最终模型在Urban100/Manga109数据集上实现了最佳性能32.97/39.24 dB PSNR。实验验证了所提方法中各组件的有效性。**注意力状态空间方程消融实验**。在第4.1节中，我们指出状态空间方程中的矩阵C实际上与注意力机制中的查询行为类似，因此我们提出通过残差加法将全局提示插入C的ASE方案。表2中，我们探索了ASE中提示的其他插入位置。由于B可以

类似于注意力机制中的Key，将提示插入B也能获得不错的结果，但略逊于添加到C的效果。我们认为这是因为C在状态空间方程等式（1）中更接近输出端，从而对最终结果产生更大影响。此外，将提示插入离散时间步长 $\Delta$ 和输出y均未能获得满意性能，这证明了我们在所提出的ASE中聚焦于矩阵C的合理性。由于篇幅限制，我们在Suppl.中提供了更多消融实验。

### 5.3. 图像超分辨率技术比较

**轻量级图像超分辨率**。继先前研究[33, 58]之后，我们同样将参数数量（#param）和MACs（将低分辨率图像放大至1280 × 720分辨率）作为效率指标进行报告。表3的结果表明，我们的MambaIRv2在参数数量显著更少的情况下，仍优于当前最先进的方法SRFormer[58]。例如，在2 × Urban100数据集上，我们的MambaIRv2-light比SRFormer-light性能提升了0.35dB。



Table 4. Quantitative comparison on *classic image super-resolution* with state-of-the-art methods.

| Method       | scale | #param | Set5  |        | Set14 |        | BSDS100 |        | Urban100 |        | Manga109 |        |
|--------------|-------|--------|-------|--------|-------|--------|---------|--------|----------|--------|----------|--------|
|              |       |        | PSNR  | SSIM   | PSNR  | SSIM   | PSNR    | SSIM   | PSNR     | SSIM   | PSNR     | SSIM   |
| EDSR [30]    | 2×    | 42.6M  | 38.11 | 0.9602 | 33.92 | 0.9195 | 32.32   | 0.9013 | 32.93    | 0.9351 | 39.10    | 0.9773 |
| RCAN [52]    | 2×    | 15.4M  | 38.27 | 0.9614 | 34.12 | 0.9216 | 32.41   | 0.9027 | 33.34    | 0.9384 | 39.44    | 0.9786 |
| SAN [12]     | 2×    | 15.7M  | 38.31 | 0.9620 | 34.07 | 0.9213 | 32.42   | 0.9028 | 33.10    | 0.9370 | 39.32    | 0.9792 |
| IPT [5]      | 2×    | 115M   | 38.37 | -      | 34.43 | -      | 32.48   | -      | 33.76    | -      | -        | -      |
| SwinIR [29]  | 2×    | 11.8M  | 38.42 | 0.9623 | 34.46 | 0.9250 | 32.53   | 0.9041 | 33.81    | 0.9427 | 39.92    | 0.9797 |
| EDT [27]     | 2×    | 11.5M  | 38.45 | 0.9624 | 34.57 | 0.9258 | 32.52   | 0.9041 | 33.80    | 0.9425 | 39.93    | 0.9800 |
| MambaIR [18] | 2×    | 20.4M  | 38.57 | 0.9627 | 34.67 | 0.9261 | 32.58   | 0.9048 | 34.15    | 0.9446 | 40.28    | 0.9806 |
| CAT-A [8]    | 2×    | 16.5M  | 38.51 | 0.9626 | 34.78 | 0.9265 | 32.59   | 0.9047 | 34.26    | 0.9440 | 40.10    | 0.9805 |
| DAT [10]     | 2×    | 14.8M  | 38.58 | 0.9629 | 34.81 | 0.9297 | 32.61   | 0.9051 | 34.37    | 0.9458 | 40.33    | 0.9807 |
| HAT [7]      | 2×    | 20.6M  | 38.63 | 0.9630 | 34.86 | 0.9274 | 32.62   | 0.9053 | 34.45    | 0.9466 | 40.26    | 0.9809 |
| MambaIRv2-S  | 2×    | 9.6M   | 38.53 | 0.9627 | 34.62 | 0.9256 | 32.59   | 0.9048 | 34.24    | 0.9454 | 40.27    | 0.9808 |
| MambaIRv2-B  | 2×    | 22.9M  | 38.65 | 0.9631 | 34.89 | 0.9275 | 32.62   | 0.9053 | 34.49    | 0.9468 | 40.42    | 0.9810 |
| MambaIRv2-L  | 2×    | 34.2M  | 38.65 | 0.9632 | 34.93 | 0.9276 | 32.62   | 0.9053 | 34.60    | 0.9475 | 40.55    | 0.9807 |
| <hr/>        |       |        |       |        |       |        |         |        |          |        |          |        |
| EDSR [30]    | 3×    | 42.6M  | 34.65 | 0.9280 | 30.52 | 0.8462 | 29.25   | 0.8093 | 28.80    | 0.8653 | 34.17    | 0.9476 |
| RCAN [52]    | 3×    | 15.4M  | 34.74 | 0.9299 | 30.65 | 0.8482 | 29.32   | 0.8111 | 29.09    | 0.8702 | 34.44    | 0.9499 |
| SAN [12]     | 3×    | 15.7M  | 34.75 | 0.9300 | 30.59 | 0.8476 | 29.33   | 0.8112 | 28.93    | 0.8671 | 34.30    | 0.9494 |
| IPT [5]      | 3×    | 115M   | 34.81 | -      | 30.85 | -      | 29.38   | -      | 29.49    | -      | -        | -      |
| SwinIR [29]  | 3×    | 11.8M  | 34.97 | 0.9318 | 30.93 | 0.8534 | 29.46   | 0.8145 | 29.75    | 0.8826 | 35.12    | 0.9537 |
| EDT [27]     | 3×    | 11.5M  | 34.97 | 0.9316 | 30.89 | 0.8527 | 29.44   | 0.8142 | 29.72    | 0.8814 | 35.13    | 0.9534 |
| MambaIR [18] | 3×    | 20.4M  | 35.08 | 0.9323 | 30.99 | 0.8536 | 29.51   | 0.8157 | 29.93    | 0.8841 | 35.43    | 0.9546 |
| CAT-A [8]    | 3×    | 16.5M  | 35.06 | 0.9326 | 31.04 | 0.8538 | 29.52   | 0.8160 | 30.12    | 0.8862 | 35.80    | 0.9546 |
| DAT [10]     | 3×    | 14.8M  | 35.16 | 0.9331 | 31.11 | 0.8550 | 29.55   | 0.8169 | 30.18    | 0.8886 | 35.59    | 0.9554 |
| HAT [7]      | 3×    | 20.6M  | 35.07 | 0.9329 | 31.08 | 0.8555 | 29.54   | 0.8167 | 30.23    | 0.8896 | 35.53    | 0.9552 |
| MambaIRv2-S  | 3×    | 9.8M   | 35.09 | 0.9326 | 31.07 | 0.8547 | 29.51   | 0.8157 | 30.08    | 0.8871 | 35.44    | 0.9549 |
| MambaIRv2-B  | 3×    | 23.1M  | 35.18 | 0.9334 | 31.12 | 0.8557 | 29.55   | 0.8169 | 30.28    | 0.8905 | 35.61    | 0.9556 |
| MambaIRv2-L  | 3×    | 34.2M  | 35.16 | 0.9334 | 31.18 | 0.8564 | 29.57   | 0.8175 | 30.34    | 0.8912 | 35.72    | 0.9561 |
| <hr/>        |       |        |       |        |       |        |         |        |          |        |          |        |
| EDSR [30]    | 4×    | 43.0M  | 32.46 | 0.8968 | 28.80 | 0.7876 | 27.71   | 0.7420 | 26.64    | 0.8033 | 31.02    | 0.9148 |
| RCAN [52]    | 4×    | 15.6M  | 32.63 | 0.9002 | 28.87 | 0.7889 | 27.77   | 0.7436 | 26.82    | 0.8087 | 31.22    | 0.9173 |
| SAN [12]     | 4×    | 15.7M  | 32.64 | 0.9003 | 28.92 | 0.7888 | 27.78   | 0.7436 | 26.79    | 0.8068 | 31.18    | 0.9169 |
| IPT [5]      | 4×    | 116M   | 32.64 | -      | 29.01 | -      | 27.82   | -      | 27.26    | -      | -        | -      |
| SwinIR [29]  | 4×    | 11.9M  | 32.92 | 0.9044 | 29.09 | 0.7950 | 27.92   | 0.7489 | 27.45    | 0.8254 | 32.03    | 0.9260 |
| EDT [27]     | 4×    | 11.6M  | 32.82 | 0.9031 | 29.09 | 0.7939 | 27.91   | 0.7483 | 27.46    | 0.8246 | 32.05    | 0.9254 |
| MambaIR [18] | 4×    | 20.4M  | 33.03 | 0.9046 | 29.20 | 0.7961 | 27.98   | 0.7503 | 27.68    | 0.8287 | 32.32    | 0.9272 |
| CAT-A [8]    | 4×    | 16.6M  | 33.08 | 0.9052 | 29.18 | 0.7960 | 27.99   | 0.7510 | 27.89    | 0.8339 | 32.39    | 0.9285 |
| DAT [10]     | 4×    | 14.8M  | 33.08 | 0.9055 | 29.23 | 0.7973 | 28.00   | 0.7515 | 27.87    | 0.8343 | 32.51    | 0.9291 |
| HAT [7]      | 4×    | 20.8M  | 33.04 | 0.9056 | 29.23 | 0.7973 | 28.00   | 0.7517 | 27.97    | 0.8368 | 32.48    | 0.9292 |
| MambaIRv2-S  | 4×    | 9.8M   | 32.99 | 0.9037 | 29.23 | 0.7965 | 27.97   | 0.7502 | 27.73    | 0.8307 | 32.33    | 0.9276 |
| MambaIRv2-B  | 4×    | 23.1M  | 33.14 | 0.9057 | 29.23 | 0.7975 | 28.00   | 0.7511 | 27.89    | 0.8344 | 32.57    | 0.9295 |
| MambaIRv2-L  | 4×    | 34.2M  | 33.19 | 0.9062 | 29.29 | 0.7982 | 28.01   | 0.7521 | 28.07    | 0.8383 | 32.66    | 0.9304 |

Table 5. Complexity comparison with state-of-the-art methods on the 2× classic SR with output size 256 × 256.

| Models      | #param | MACs   | Urban100 |        | Manga109 |        |
|-------------|--------|--------|----------|--------|----------|--------|
|             |        |        | PSNR     | SSIM   | PSNR     | SSIM   |
| CAT-A [8]   | 16.6M  | 350.7G | 34.26    | 0.9440 | 40.10    | 0.9805 |
| DAT [10]    | 14.8M  | 265.7G | 34.37    | 0.9458 | 40.33    | 0.9807 |
| HAT [7]     | 20.8M  | 514.9G | 34.45    | 0.9466 | 40.26    | 0.9809 |
| MambaIRv2-S | 9.6M   | 192.9G | 34.24    | 0.9454 | 40.27    | 0.9808 |
| MambaIRv2-B | 22.9M  | 445.8G | 34.49    | 0.9468 | 40.42    | 0.9810 |
| MambaIRv2-L | 34.2M  | 664.5G | 34.60    | 0.9475 | 40.55    | 0.9807 |

79K fewer #param. This experiment validates the efficiency and effectiveness of the proposed method.

**Classic Image Super-Resolution.** Tab. 4 gives the comparison results of our MambaIRv2 with different model sizes to existing state-of-the-art classic SR methods. Thanks to the attentive state space modeling, our proposed method achieves the best performance across most five benchmark datasets and up-sample scales. For example, our MambaIRv2-B exceeds HAT [7] by 0.16dB on the 2× Manga109 dataset. Interestingly, even the MambaIRv2-S

with 9.6M #param, outperforms the previous 20.4M MambaIR [18] by 0.09dB PSNR on the 2× Urban100 dataset, further demonstrating our MambaIRv2 serves as an elegant balance of performance and efficiency. Finally, our method also shows promising scaling-up capabilities. When we scale up the #param to 34.2M to obtain the MambaIRv2-L model, this larger model achieved steady performance gains compared to its base counterpart, *e.g.*, 0.18dB PSNR gains on 4× Urban100. The visual comparisons are shown in Fig. 4, and our method can facilitate the reconstruction of sharp edges and natural textures.

**Model Complexity Comparison.** As shown in Tab. 5, our MambaIRv2-S model, which uses only 55.0% of the MACs compared to CAT-A [8], outperforms CAT-A by 0.17dB PSNR on Manga109. Additionally, our MambaIRv2-B model, which roughly matches the #param of HAT [7], achieves a 13.4% reduction in MACs, while delivering 0.04/0.16dB PSNR improvements on Urban100/Manga109. The above results demonstrate that our MambaIRv2 strikes a sweet spot between performance and efficiency.

表4. 经典图像超分辨率与最先进方法的定量比较。

| 方法           | 规模 | #参数   | Set5     |        | Set14    |        | BSDS100  |        | 城市100    |        | Manga109 |        |
|--------------|----|-------|----------|--------|----------|--------|----------|--------|----------|--------|----------|--------|
|              |    |       | 信号-噪音功率比 | SSIM   | 信号-噪音功率比 | SSIM   | 信号-噪音功率比 | SSIM   | 信号-噪音功率比 | SSIM   | 信号-噪音功率比 | SSIM   |
| EDSR [30]    | 2× | 42.6M | 38.11    | 0.9602 | 33.92    | 0.9195 | 32.32    | 0.9013 | 32.93    | 0.9351 | 39.10    | 0.9773 |
| RCAN [52]    | 2× | 15.4M | 38.27    | 0.9614 | 34.12    | 0.9216 | 32.41    | 0.9027 | 33.34    | 0.9384 | 39.44    | 0.9786 |
| SAN [12]     | 2× | 15.7M | 38.31    | 0.9620 | 34.07    | 0.9213 | 32.42    | 0.9028 | 33.10    | 0.9370 | 39.32    | 0.9792 |
| IPT [5]      | 2× | 115M  | 38.37    | -      | 34.43    | -      | 32.48    | -      | 33.76    | -      | -        | -      |
| SwinIR [29]  | 2× | 11.8M | 38.42    | 0.9623 | 34.46    | 0.9250 | 32.53    | 0.9041 | 33.81    | 0.9427 | 39.92    | 0.9797 |
| EDT [27]     | 2× | 11.5M | 38.45    | 0.9624 | 34.57    | 0.9258 | 32.52    | 0.9041 | 33.80    | 0.9425 | 39.93    | 0.9800 |
| MambaIR [18] | 2× | 20.4M | 38.57    | 0.9627 | 34.67    | 0.9261 | 32.58    | 0.9048 | 34.15    | 0.9446 | 40.28    | 0.9806 |
| CAT-A [8]    | 2× | 16.5M | 38.51    | 0.9626 | 34.78    | 0.9265 | 32.59    | 0.9047 | 34.26    | 0.9440 | 40.10    | 0.9805 |
| DAT [10]     | 2× | 14.8M | 38.58    | 0.9629 | 34.81    | 0.9297 | 32.61    | 0.9051 | 34.37    | 0.9458 | 40.33    | 0.9807 |
| HAT [7]      | 2× | 20.6M | 38.63    | 0.9630 | 34.86    | 0.9274 | 32.62    | 0.9053 | 34.45    | 0.9466 | 40.26    | 0.9809 |
| MambaIRv2-S  | 2× | 9.6M  | 38.53    | 0.9627 | 34.62    | 0.9256 | 32.59    | 0.9048 | 34.24    | 0.9454 | 40.27    | 0.9808 |
| MambaIRv2-B  | 2× | 22.9M | 38.65    | 0.9631 | 34.89    | 0.9275 | 32.62    | 0.9053 | 34.49    | 0.9468 | 40.42    | 0.9810 |
| MambaIRv2-L  | 2× | 34.2M | 38.65    | 0.9632 | 34.93    | 0.9276 | 32.62    | 0.9053 | 34.60    | 0.9475 | 40.55    | 0.9807 |
| EDSR [30]    | 3× | 42.6M | 34.65    | 0.9280 | 30.52    | 0.8462 | 29.25    | 0.8093 | 28.80    | 0.8653 | 34.17    | 0.9476 |
| RCAN [52]    | 3× | 15.4M | 34.74    | 0.9299 | 30.65    | 0.8482 | 29.32    | 0.8111 | 29.09    | 0.8702 | 34.44    | 0.9499 |
| SAN [12]     | 3× | 15.7M | 34.75    | 0.9300 | 30.59    | 0.8476 | 29.33    | 0.8112 | 28.93    | 0.8671 | 34.30    | 0.9494 |
| IPT [5]      | 3× | 115M  | 34.81    | -      | 30.85    | -      | 29.38    | -      | 29.49    | -      | -        | -      |
| SwinIR [29]  | 3× | 11.8M | 34.97    | 0.9318 | 30.93    | 0.8534 | 29.46    | 0.8145 | 29.75    | 0.8826 | 35.12    | 0.9537 |
| EDT [27]     | 3× | 11.5M | 34.97    | 0.9316 | 30.89    | 0.8527 | 29.44    | 0.8142 | 29.72    | 0.8814 | 35.13    | 0.9534 |
| MambaIR [18] | 3× | 20.4M | 35.08    | 0.9323 | 30.99    | 0.8536 | 29.51    | 0.8157 | 29.93    | 0.8841 | 35.43    | 0.9546 |
| CAT-A [8]    | 3× | 16.5M | 35.06    | 0.9326 | 31.04    | 0.8538 | 29.52    | 0.8160 | 30.12    | 0.8862 | 35.80    | 0.9546 |
| DAT [10]     | 3× | 14.8M | 35.16    | 0.9331 | 31.11    | 0.8550 | 29.55    | 0.8169 | 30.18    | 0.8886 | 35.59    | 0.9554 |
| HAT [7]      | 3× | 20.6M | 35.07    | 0.9329 | 31.08    | 0.8555 | 29.54    | 0.8167 | 30.23    | 0.8896 | 35.53    | 0.9552 |
| MambaIRv2-S  | 3× | 9.8M  | 35.09    | 0.9326 | 31.07    | 0.8547 | 29.51    | 0.8157 | 30.08    | 0.8871 | 35.44    | 0.9549 |
| MambaIRv2-B  | 3× | 23.1M | 35.18    | 0.9334 | 31.12    | 0.8557 | 29.55    | 0.8169 | 30.28    | 0.8905 | 35.61    | 0.9556 |
| MambaIRv2-L  | 3× | 34.2M | 35.16    | 0.9334 | 31.18    | 0.8564 | 29.57    | 0.8175 | 30.34    | 0.8912 | 35.72    | 0.9561 |
| EDSR [30]    | 4× | 43.0M | 32.46    | 0.8968 | 28.80    | 0.7876 | 27.71    | 0.7420 | 26.64    | 0.8033 | 31.02    | 0.9148 |
| RCAN [52]    | 4× | 15.6M | 32.63    | 0.9002 | 28.87    | 0.7889 | 27.77    | 0.7436 | 26.82    | 0.8087 | 31.22    | 0.9173 |
| SAN [12]     | 4× | 15.7M | 32.64    | 0.9003 | 28.92    | 0.7888 | 27.78    | 0.7436 | 26.79    | 0.8068 | 31.18    | 0.9169 |
| IPT [5]      | 4× | 116M  | 32.64    | -      | 29.01    | -      | 27.82    | -      | 27.26    | -      | -        | -      |
| SwinIR [29]  | 4× | 11.9M | 32.92    | 0.9044 | 29.09    | 0.7950 | 27.92    | 0.7489 | 27.45    | 0.8254 | 32.03    | 0.9260 |
| EDT [27]     | 4× | 11.6M | 32.82    | 0.9031 | 29.09    | 0.7939 | 27.91    | 0.7483 | 27.46    | 0.8246 | 32.05    | 0.9254 |
| MambaIR [18] | 4× | 20.4M | 33.03    | 0.9046 | 29.20    | 0.7961 | 27.98    | 0.7503 | 27.68    | 0.8287 | 32.32    | 0.9272 |
| CAT-A [8]    | 4× | 16.6M | 33.08    | 0.9052 | 29.18    | 0.7960 | 27.99    | 0.7510 | 27.89    | 0.8339 | 32.39    | 0.9285 |
| DAT [10]     | 4× | 14.8M | 33.08    | 0.9055 | 29.23    | 0.7973 | 28.00    | 0.7515 | 27.87    | 0.8343 | 32.51    | 0.9291 |
| HAT [7]      | 4× | 20.8M | 33.04    | 0.9056 | 29.23    | 0.7973 | 28.00    | 0.7517 | 27.97    | 0.8368 | 32.48    | 0.9292 |
| MambaIRv2-S  | 4× | 9.8M  | 32.99    | 0.9037 | 29.23    | 0.7965 | 27.97    | 0.7502 | 27.73    | 0.8307 | 32.33    | 0.9276 |
| MambaIRv2-B  | 4× | 23.1M | 33.14    | 0.9057 | 29.23    | 0.7975 | 28.00    | 0.7511 | 27.89    | 0.8344 | 32.57    | 0.9295 |
| MambaIRv2-L  | 4× | 34.2M | 33.19    | 0.9062 | 29.29    | 0.7982 | 28.01    | 0.7521 | 28.07    | 0.8383 | 32.66    | 0.9304 |

表5. 在输出尺寸为256 × 256的2 ×经典SR数据集上与最先进方法的复杂度比较。

| 模型          | #参数   | 多通道控制器 | 城市100    |        | Manga109 |        |
|-------------|-------|--------|----------|--------|----------|--------|
|             |       |        | 信号-噪音功率比 | SSIM   | 信号-噪音功率比 | SSIM   |
| CAT-A [8]   | 16.6兆 | 350.7吉 | 34.26    | 0.9440 | 40.10    | 0.9805 |
| DAT [10]    | 14.8M | 265.7G | 34.37    | 0.9458 | 40.33    | 0.9807 |
| HAT [7]     | 20.8M | 514.9G | 34.45    | 0.9466 | 40.26    | 0.9809 |
| MambaIRv2-S | 9.6M  | 192.9G | 34.24    | 0.9454 | 40.27    | 0.9808 |
| MambaIRv2-B | 22.9M | 445.8G | 34.49    | 0.9468 | 40.42    | 0.9810 |
| MambaIRv2-L | 34.2M | 664.5G | 34.60    | 0.9475 | 40.55    | 0.9807 |

参数数量减少79K个。本实验验证了所提出方法的效率与有效性。

**经典图像超分辨率。**表4展示了我们的MambaIRv2在不同模型规模下与现有最先进经典SR方法的对比结果。得益于注意力状态空间建模，我们提出的方法在大多数五个基准数据集和上采样尺度上均取得最佳性能。例如，在2 × Manga109数据集上，我们的MambaIRv2-B比HAT[7]高出0.16dB。值得注意的是，即使MambaIRv2-S

当参数量为960万时，在2 × Urban100数据集上表现优于先前2040万参数量的MambaIR[18]，PSNR 提升达0.09dB，进一步证明我们的MambaIRv2在性能与效率之间实现了优雅平衡。最后，该方法还展现出良好的扩展能力：当我们将参数量扩展至3420万以构建MambaIRv2-L模型时，相较于基础模型，该大型模型在4 × Urban100数据集上实现了稳定的性能提升（例如0.18dB的PSNR增益）。如图4所示，我们的方法能够有效重建锐利边缘与自然纹理。

**模型复杂度比较。**如表5所示，我们的MambaIRv2-S模型仅使用了MACs的55.0%（相较于CAT-A[8]），在Manga109数据集上仍比CAT-A高出0.17dB的PSNR。此外，我们的MambaIRv2-B模型参数量与HAT[7]大致相当，MACs减少了13.4%，同时在Urban100/Manga109数据集上实现了0.04/0.16dB的PSNR提升。上述结果表明，我们的MambaIRv2在性能与效率之间找到了最佳平衡点。

Table 6. Quantitative comparison on *JPEG compression artifact reduction* under different quality factors  $q$ .

| Dataset  | $q$ | RNAN [54] |        | RDN [55] |        | DRUNet [48] |        | SwinIR [29] |        | MambaIR [18] |               | Ours         |               |
|----------|-----|-----------|--------|----------|--------|-------------|--------|-------------|--------|--------------|---------------|--------------|---------------|
|          |     | PSNR      | SSIM   | PSNR     | SSIM   | PSNR        | SSIM   | PSNR        | SSIM   | PSNR         | SSIM          | PSNR         | SSIM          |
| Classic5 | 10  | 29.96     | 0.8178 | 30.00    | 0.8188 | 30.16       | 0.8234 | 30.27       | 0.8249 | <b>30.27</b> | <b>0.8256</b> | <b>30.37</b> | <b>0.8269</b> |
|          | 30  | 33.38     | 0.8924 | 33.43    | 0.8930 | 33.59       | 0.8949 | 33.73       | 0.8961 | <b>33.74</b> | <b>0.8965</b> | <b>33.81</b> | <b>0.8970</b> |
|          | 40  | 34.27     | 0.9061 | 34.27    | 0.9061 | 34.41       | 0.9075 | 34.52       | 0.9082 | <b>34.53</b> | <b>0.9084</b> | <b>34.64</b> | <b>0.9093</b> |
| LIVE1    | 10  | 29.63     | 0.8239 | 29.67    | 0.8247 | 29.79       | 0.8278 | 29.86       | 0.8287 | <b>29.88</b> | <b>0.8301</b> | <b>29.91</b> | <b>0.8301</b> |
|          | 30  | 33.45     | 0.9149 | 33.51    | 0.9153 | 33.59       | 0.9166 | 33.69       | 0.9174 | <b>33.72</b> | <b>0.9179</b> | <b>33.73</b> | <b>0.9179</b> |
|          | 40  | 34.47     | 0.9299 | 34.51    | 0.9302 | 34.58       | 0.9312 | 34.67       | 0.9317 | <b>34.70</b> | <b>0.9320</b> | <b>34.73</b> | <b>0.9323</b> |

Table 7. Quantitative comparison of PSNR on *gaussian color image denoising*  $\sigma = 15$  with state-of-the-art methods.

| Method         | CBSD68       | Kodak24      | McMaster     | Urban100     |
|----------------|--------------|--------------|--------------|--------------|
| IRCNN [46]     | 33.86        | 34.69        | 34.58        | 33.78        |
| FFDNet [47]    | 33.87        | 34.63        | 34.66        | 33.83        |
| DnCNN [45]     | 33.90        | 34.60        | 33.45        | 32.98        |
| DRUNet [48]    | 34.30        | 35.31        | 35.40        | 34.81        |
| SwinIR [29]    | 34.42        | 35.34        | 35.61        | 35.13        |
| Restormer [42] | 34.40        | 35.35        | 35.61        | 35.13        |
| Xformer [43]   | 34.43        | 35.39        | 35.68        | 35.29        |
| MambaIR [18]   | <b>34.48</b> | <b>35.42</b> | <b>35.70</b> | <b>35.37</b> |
| MambaIRv2      | <b>34.48</b> | <b>35.43</b> | <b>35.73</b> | <b>35.42</b> |

Table 8. Comparison to MambaIR with different scanning modes on  $2\times$  lightweight SR. The “MambaIR- $n$ ” indicates the MambaIR [18] variant with  $n$  scanning directions.

| settings       | #param      | MACs        | Urban100     |               | Manga109     |               |
|----------------|-------------|-------------|--------------|---------------|--------------|---------------|
|                |             |             | PSNR         | SSIM          | PSNR         | SSIM          |
| MambaIR-1 [18] | 987K        | 291G        | 32.82        | 0.9348        | 39.19        | 0.9776        |
| MambaIR-2 [18] | 1.11M       | 383G        | 32.86        | 0.9450        | 39.26        | 0.9778        |
| MambaIR-4 [18] | 1.36M       | 568G        | 32.92        | 0.9356        | 39.31        | 0.9779        |
| MambaIRv2      | <b>774K</b> | <b>286G</b> | <b>33.26</b> | <b>0.9378</b> | <b>39.39</b> | <b>0.9786</b> |

## 5.4. Comparison on JPEG CAR

Tab. 6 gives the results on JPEG compression reduction. It can be seen that the proposed MambaIRv2 achieves the best performance on all testing datasets across all quality factors. For example, our MambaIRv2 suppresses MambaIR [18] by 0.11dB PSNR with  $q = 40$  on the Classic5 dataset, demonstrating the effectiveness of our MambaIRv2 on other restoration tasks.

## 5.5. Comparison on Image Denoising

We further include the gaussian color image denoising task for further validation. Notably, to maintain architectural consistency across different restoration tasks, we retain the straight-through structure and avoid using the UNet architecture, which has been shown to be particularly advantageous for denoising tasks [6]. The results presented in Tab. 7 demonstrate that MambaIRv2 outperforms all other models across the datasets. In particular, it surpasses U-shaped Restormer [42] by even 0.29dB PSNR on the Urban100 dataset. This experiment validates our MambaIRv2 serves as a generalized image restoration backbone.

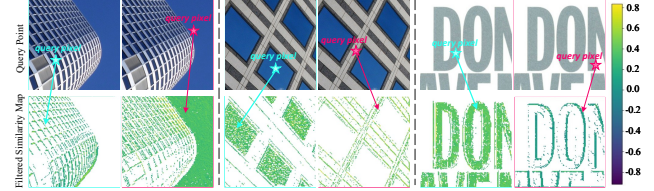


Figure 5. The visualization of the attentive state space. We compute the cosine similarity between the prompt corresponding to the query pixel and the matrix  $C$ . We filter out low-similarity points for presentation clarity. More examples are provided in the *Suppl.*

## 5.6. Discussion

**Benefits from Reduced Scan Directions.** Compared to the previous MambaIR [18], which performs 4 scans in pixel space, a significant advantage of our MambaIRv2 is that it requires only a single scan in the semantic space. As shown in Tab. 8, our MambaIRv2 is not only efficient but also boosts performance. For example, MambaIRv2 reduces even **43%** of the number of parameters and **50%** computational burden compared to the standard MambaIR, while still suppresses by **0.34dB** PSNR on  $2\times$  Urban100.

**Visualization of Attentive State Space.** In the proposed attentive state space equation, the prompts play an important role in representing similar pixels across the whole image to facilitate the query pixel seeing out of the scanned sequence. As shown in Fig. 5, it can be seen that the query pixel is empowered to attend to its corresponding semantic part in the image through the prompt, thus enabling global information aggregation similar to attention mechanism.

## 6. Conclusion

In this work, we introduce MambaIRv2 to enhance state-space restoration models by addressing the causal modeling nature of Mamba. We propose the attentive state-space equation that incorporates prompt learning for enlarged token perception as well as scanning only once. Additionally, we introduce semantic guided neighboring which positions similar pixels closer to handle the long-range decay. These innovations enable MambaIRv2 to integrate ViT-like non-causal abilities into Mamba-based models to implement the attentive state space restoration. Extensive experiments confirm our MambaIRv2 as an efficient, high-performing backbone for image restoration.



表6. 不同质量因子 $q$ 下JPEG压缩伪影减少的定量比较。

| 数据集   | $q$ | RNAN [54] |        | RDN [55] |        | DRUNet [48] |        | SwinIR [29] |        | MambaIR [18] |        | 我们的      |        |
|-------|-----|-----------|--------|----------|--------|-------------|--------|-------------|--------|--------------|--------|----------|--------|
|       |     | 信号-噪声功率比  | SSIM   | 信号-噪声功率比 | SSIM   | 信号-噪声功率比    | SSIM   | 信号-噪声功率比    | SSIM   | 信号-噪声功率比     | SSIM   | 信号-噪声功率比 | SSIM   |
| 经典5   | 10  | 29.96     | 0.8178 | 30.00    | 0.8188 | 30.16       | 0.8234 | 30.27       | 0.8249 | 30.27        | 0.8256 | 30.37    | 0.8269 |
|       | 30  | 33.38     | 0.8924 | 33.43    | 0.8930 | 33.59       | 0.8949 | 33.73       | 0.8961 | 33.74        | 0.8965 | 33.81    | 0.8970 |
|       | 40  | 34.27     | 0.9061 | 34.27    | 0.9061 | 34.41       | 0.9075 | 34.52       | 0.9082 | 34.53        | 0.9084 | 34.64    | 0.9093 |
| LIVE1 | 10  | 29.63     | 0.8239 | 29.67    | 0.8247 | 29.79       | 0.8278 | 29.86       | 0.8287 | 29.88        | 0.8301 | 29.91    | 0.8301 |
|       | 30  | 33.45     | 0.9149 | 33.51    | 0.9153 | 33.59       | 0.9166 | 33.69       | 0.9174 | 33.72        | 0.9179 | 33.73    | 0.9179 |
|       | 40  | 34.47     | 0.9299 | 34.51    | 0.9302 | 34.58       | 0.9312 | 34.67       | 0.9317 | 34.70        | 0.9320 | 34.73    | 0.9323 |

表7. 基于高斯彩色图像去噪的 PSNR 与 $\xi=15$ 时与最先进方法的定量比较。

| 方法             | CBSD68 | 柯达24  | 麦克马斯特大学测试集的英国测试 | 城市100 |
|----------------|--------|-------|-----------------|-------|
| IRCNN [46]     | 33.86  | 34.69 | 34.58           | 33.78 |
| FFDNet [47]    | 33.87  | 34.63 | 34.66           | 33.83 |
| DnCNN [45]     | 33.90  | 34.60 | 33.45           | 32.98 |
| DRUNet [48]    | 34.30  | 35.31 | 35.40           | 34.81 |
| SwinIR [29]    | 34.42  | 35.34 | 35.61           | 35.13 |
| Restormer [42] | 34.40  | 35.35 | 35.61           | 35.13 |
| Xformer [43]   | 34.43  | 35.39 | 35.68           | 35.29 |
| MambaIR [18]   | 34.48  | 35.42 | 35.70           | 35.37 |
| MambaIRv2      | 34.48  | 35.43 | 35.73           | 35.42 |

表8. 在 $2 \times$ 轻量级SR上采用不同扫描模式与MambaIR的对比。

“MambaIR-n” 表示具有 $n$ 个扫描方向的MambaIR[18]变体。

| 镶嵌             | #参数   | MAC地址 | 城市100 |        | Manga109 |        |
|----------------|-------|-------|-------|--------|----------|--------|
|                |       |       | PSNR  | SSIM   | PSNR     | SSIM   |
| MambaIR-1 [18] | 987K  | 291G  | 32.82 | 0.9348 | 39.19    | 0.9776 |
| MambaIR-2 [18] | 1.11M | 383G  | 32.86 | 0.9450 | 39.26    | 0.9778 |
| MambaIR-4 [18] | 1.36M | 568G  | 32.92 | 0.9356 | 39.31    | 0.9779 |
| MambaIRv2      | 774K  | 286G  | 33.26 | 0.9378 | 39.39    | 0.9786 |

## 5.4. JPEG与CAR格式对比研究

表6展示了JPEG压缩率降低的结果。可以看出，所提出的MambaIRv2在所有测试数据集上均实现了最佳性能，且不受质量因子影响。例如，在Classic5数据集上，当 $q=40$ 时，我们的MambaIRv2将MambaIR[18]的压缩率降低了0.11dB PSNR，这证明了MambaIRv2在其他图像恢复任务中的有效性。

## 5.5. 图像去噪方法比较

我们进一步加入了高斯颜色图像去噪任务以进行更全面的验证。值得注意的是，为确保不同修复任务间架构的一致性，我们保留了直通式结构并避免使用UNet架构——已有研究表明该架构在去噪任务中具有显著优势[6]。表7所示结果表明，MambaIRv2在所有数据集上均优于其他模型。特别是在Urban100数据集上，其表现甚至比U型Restormer[42]高出0.29dB的PSNR。该实验验证了MambaIRv2可作为通用图像修复骨干网络。

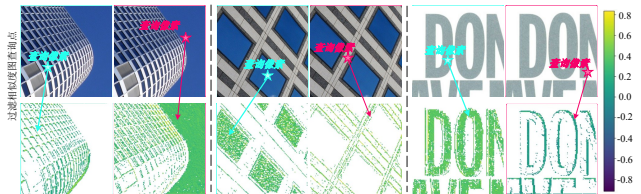


图5. 注意力状态空间的可视化。我们计算查询像素对应的提示与矩阵 $C$ 之间的余弦相似度。为呈现清晰度，我们过滤了低相似度点。更多示例见补充材料。

## 5.6. 讨论

**降低扫描方向带来的优势。**与先前在像素空间执行4次扫描的MambaIR[18]相比，我们MambaIRv2的一个显著优势在于仅需在语义空间进行单次扫描。如表8所示，MambaIRv2不仅高效，还能提升性能。例如，在 $2 \times$  Urban100数据集上，相较于标准MambaIR，MambaIRv2甚至能减少43%的参数量和50%的计算负担，同时仍能抑制0.34dB的PSNR。注意力状态空间可视化。在提出的注意力状态空间方程中，提示词在表征整个图像中相似像素方面起关键作用，从而帮助查询像素跳出扫描序列。如图5所示，查询像素通过提示词被赋予关注图像中对应语义部分的能力，从而实现类似注意力机制的全局信息聚合。

## 6. 结论

在本研究中，我们提出MambaIRv2框架，通过解决Mamba模型的因果建模特性来增强状态空间恢复模型。我们创新性地设计了注意力状态空间方程，该方程融合了提示学习机制以提升标记感知能力，并采用单次扫描策略。此外，我们引入语义引导邻域算法，通过将相似像素位置更紧密地关联来有效处理长程衰减问题。这些创新设计使MambaIRv2能够将类似视觉信息图卷积器（ViT）的非因果能力整合到基于Mamba的模型中，从而实现注意力状态空间恢复功能。大量实验验证表明，MambaIRv2作为图像恢复模型的骨干网络展现出高效能与高精度的双重优势。

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